

Designing Inclusive Digital Solutions

An Open-Source UI/UX Guide for Designing Context-Aware and Gender-Responsive Digital Solutions in Ghana

BASED ON THE PROJECT

User Testing and Support to the Adaptation of Digital Solutions for Informal Entrepreneurs



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ACCRA
2026



Implemented by:

giz Deutsche Gesellschaft
für Internationale
Zusammenarbeit (GIZ) GmbH



Imprint

Published by: Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH, under the Digital Transformation for Inclusive Entrepreneurship in Ghana (DTEG) programme.

Implemented by: GFA Consulting Group GmbH.

Guide edition: First Edition, August 2026, covering Phase 1 and Phase 2 of the User Testing and Support to the Adaptation of Digital Solutions for Informal Entrepreneurs project, 2023–2026.

Ongoing maintenance: The Ghana Hubs Network (GHN) is the intended long-term host of this guide as part of the project's sustainability plan (see Section 9).

Anonymization notice: To protect participant, intermediary, and solution-provider confidentiality, this edition generalizes and anonymizes examples throughout. Intermediary organizations and the solutions they tested are referred to by consistent reference code rather than by name; see Section 1.3 for the coding scheme.

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How to cite this guide: GIZ and GFA Consulting Group GmbH (2026). *Designing Inclusive Digital Solutions: An Open-Source UI/UX Guide for Designing Context-Aware and Gender-Responsive Digital Solutions in Ghana*. First Edition. Accra: Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH and GFA Consulting Group GmbH. Licensed under CC BY 4.0. <https://os-guide-site.vercel.app/>

Acknowledgements

This guide would not exist without the fieldwork, feedback, and trust of a great many people and organizations across Ghana.

Our thanks go to the fourteen intermediary organizations who carried out recruitment, training, co-creation facilitation, and field validation across six regions of Ghana over three years. They are identified by reference code throughout this guide to protect confidentiality, but their contribution is the evidentiary backbone of everything in these pages. See [Section 1.3](#) for the coding scheme, and the List of Contributors at the end of this guide.

We thank the thirty solution-provider teams, across both Phase 1 (adapting existing products) and Phase 2 (building new ones), who opened their products, their assumptions, and often their business models to rigorous, sometimes uncomfortable, user testing.

Above all, we thank the hundreds of women micro-entrepreneurs across Ashanti, Eastern, Northern, North East, Upper East, and Upper West regions who gave their time, their honesty, and their trust to this project, usually without any direct benefit to themselves beyond the hope that the next version of these tools would work better for people like them.

This work was commissioned by the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH and implemented by GFA Consulting Group GmbH.

Disclaimer

The findings, recommendations, and interpretations in this guide reflect the Project's own field research and the professional judgement of its implementation team. They do not necessarily represent the official position of GIZ or the German Federal Government.

This guide generalizes and anonymizes examples to protect participant, intermediary, and solution-provider confidentiality; reference codes replace names throughout. Depth of evidence varies by solution: Section [5.7](#) sets out exactly which findings carry full field validation and which are documented at roster level only, and should be read before treating any single case study as representative.

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About This Guide

The Open-Source Guide is the final output of more than three years of work done on the project **User Testing and Support to the Adaptation of Digital Solutions for Informal Micro-Entrepreneurs (2023–2026)**. The Project launched by GIZ was implemented under the Digital Transformation for Inclusive Entrepreneurship in Ghana (DTEG) programme to promote inclusive digital transformation by ensuring that digital solutions are designed around the needs of informal micro-entrepreneurs, particularly women-led enterprises. Recognizing that many existing digital products fail to address the realities of informal businesses - including low digital literacy, limited connectivity, affordability constraints, and cultural context - the project adopted a user-centred design approach to improve the relevance, accessibility, and adoption of digital technologies.

This guide translates strategies and insights from field-based user research, usability testing, innovation and co-creation activities conducted under the project into practical guidance for those designing or adapting digital solutions for informal sector entrepreneurs in Ghana. It synthesizes learnings from across the project, covering the Upper East, Northern, Eastern, and Ashanti regions of Ghana in Phase 1, and extending into Upper West and North-East Region in Phase 2. It draws on validation reports, co-creation records, Open-Source Guide input submissions, field observations, and direct contributions from all participating intermediary organizations.

All findings and recommendations presented in this guide trace directly to documented project activity. To protect participant, intermediary, and solution-provider privacy ahead of open-source publication, this edition generalizes and anonymizes examples throughout: no individual participant, staff member, or founder is named, and intermediary organizations and the solutions they tested are referred to by consistent reference codes (for example, “Intermediary N1” and “Solution S1”) rather than by name. Region, sector, and the substance of each finding are preserved in full; only identifying names are removed. Section [1.3](#) explains the coding scheme.

The guide is intended to be practical, honest, and specific. It does not summarize international design literature. It reports what happened in the field: what worked, what did not, and why.

How to Use This Document

IF YOU ARE...	START WITH	THEN READ
A hub or intermediary new to running user testing	Section 1 and Section 2	Section 4 , Part A (Steps 1–3); Section 6
A hub considering offering user testing as a paid service	Section 8	Section 1 ; Section 5 ; Section 7
A startup or solution provider preparing for user testing	Section 2 and Section 3	Section 4 , Part A or Part B depending on product stage; Section 6
A designer or developer adapting an existing product	Section 3 and Section 6	Section 4 , Part A; Section 7
A GIZ, donor, or government stakeholder reviewing project results	Key Findings at a Glance, then Section 1 and Section 5	Section 7 ; Section 9
Someone designing a new digital product from the ground up	Section 2 and Section 3	Section 4 , Part B; Section 6
Anyone wanting to contribute to or maintain this guide	Section 9	n/a

Sections [3](#), [6](#), and [7](#) are written as standing references. Return to them at any stage of a project, not only when the guide’s narrative order reaches them. Throughout the guide, colour-coded boxes flag three kinds of content: green boxes hold practical tips, blue boxes hold testimonials and observations from the field, and amber boxes flag common mistakes to avoid.

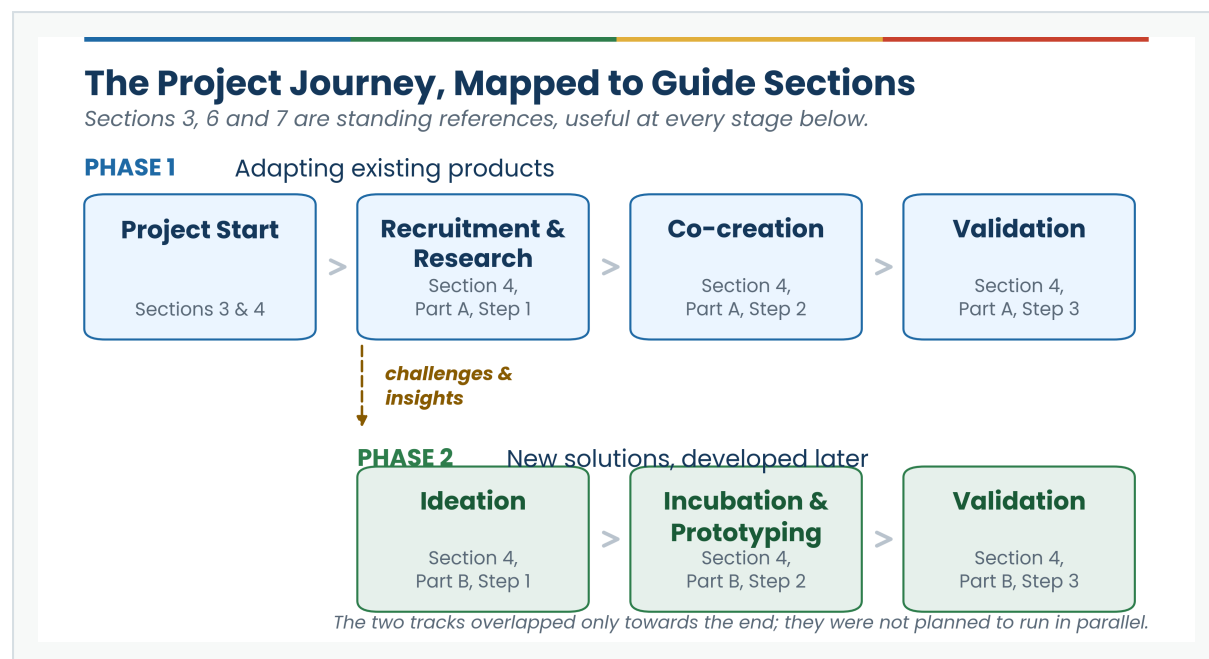


Figure 1 The project journey, mapped to guide sections

Key Findings at a Glance

Below is a summary of the key findings from the guide for a general read before diving into the full sections.

- **Trust, not technical sophistication, was the single biggest driver of adoption** in both phases. A small, verifiable promise (a fixed password flow, a specific booking time) did more for uptake than any amount of feature depth (Section [5.6](#)).
- **Language and literacy were the highest-leverage fixes available.** Solutions tested in participants' first language consistently produced more honest feedback and higher task completion than English-only equivalents (Section [5.2](#)).
- **Small, evidence-backed changes consistently beat large redesigns.** The clearest documented improvement in the project came from one targeted fix, not a rebuild (Section [5.4](#)).
- **Accessibility fixes helped everyone, not just the group they targeted.** Every accessibility improvement documented in the project (larger text, simplified navigation, voice guidance) measurably benefited users with no stated accessibility need (Sections [3.3](#), [5.5](#)).
- **Co-creation only produced measurable improvement, where solution providers had the capacity to act on findings.** Provider readiness turned out to matter as much as the quality of the research itself (Sections [5.4](#), [5.7](#)).
- **The same underlying principles held across two very different tracks:** adapting eighteen existing products (Phase 1) and validating twelve new ideas before they fully existed (Phase 2), despite different constraints, pace, and starting points (Section [5.6](#)).

Quick Start: Running Your First User Test in 5 Days

If this is your first time running a test and you do not have time to read the full guide first, start here. This gets you through one complete test. Come back to the full section once you've run a session; the details will make more sense with a real test behind you.

Before you start, check you have:

- ☐ Access to the solution or prototype idea you're testing (working login, test account, or clickable prototype)
- ☐ At least 5 people who match your target user profile: not staff, not friends, not anyone already familiar with the product
- ☐ A quiet space to run sessions, in person or by phone
- ☐ A way to take notes during the session (paper, phone, laptop, whatever you'll actually use consistently)
- ☐ Permission from each participant to take notes on the session (see the Screening Template for the wording)

If any of these are missing, get them in place before scheduling anything; a rushed test without them produces data you can't trust.

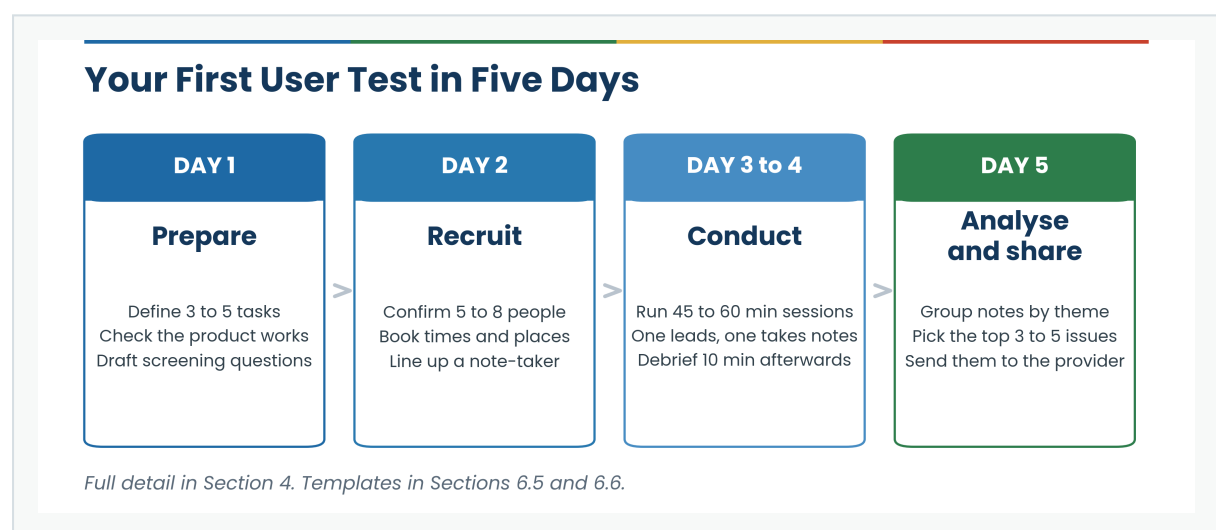


Figure 2 The five-day path to your first user test.

DAY	FOCUS	KEY ACTIONS	GO DEEPER
Day 1	Prepare	Define 3–5 concrete tasks you want participants to attempt. Confirm the solution/prototype works end to end. Write your screening questions.	Section 4 , Part A/B, Step 1; Screening Template
Day 2	Recruit and set logistics	Screen and confirm 5–8 participants (expect some no-shows). Book times, locations, and a note-taker for each session. Translate your script into the languages your participants actually speak.	Section 2.1 ; Section 6.7 (Localization)
Day 3–4	Conduct sessions	Run 45–60 minute sessions. One person facilitates, one takes notes; don't try to do both alone if you can help it. Record what participants do and say; don't explain, correct, or interpret during the session. Debrief with your team for 10 minutes after each session while it's fresh.	Section 4 , Step 2; Observation Sheet
Day 5	Analyze and share	Group your notes by task and by theme. Identify the 3–5 issues that came up most often or caused the most difficulty. Write these up in plain language and share them with the solution provider. Don't let findings sit in a notebook.	Section 4 , Step 3

COMMON MISTAKES

that undermine a first test. Testing with the wrong people (colleagues, friends, or anyone who already knows how the product works, or who may hesitate to provide honest feedback for fear of disappointing you) will not reflect a real new user. Leading questions (“Was that easy to use?” instead of “Walk me through what you’d do next”) get you agreement, not information. Skipping the debrief lets details fade fast, when a 10-minute team debrief right after each session would have captured far more. And running a test without sharing what you found has no impact, however well it was run: a finding that never reaches the solution provider changed nothing.

This Quick Start deliberately leaves out screening rigor, bias mitigation, and the differences between testing an existing product (Part A) versus a brand-new one (Part B). Section [4](#) covers all of that in full once you’re ready for it.

01

Introduction and Background

Why this guide exists, who it is for, and the project behind it.

1.1 Who This Guide Is For

Four primary audiences will find this guide most useful:

- **Innovation hubs, business coaches, and intermediary staff** advising women-led MSMEs on the adoption and use of digital solutions in peri-urban regions, including hubs in the Ghana Hubs Network considering user testing as a service line of their own.
- **Digital Solution Providers, Designers and developers** building or adapting digital solutions who want to understand the specific constraints and expectations of this user group.
- **Intermediaries supporting user research and technology adoption** in their communities, particularly those preparing to run user testing activities for the first time or improving an existing approach.
- **Development agencies and government stakeholders** assessing the project's outcomes and considering how to replicate or extend this approach will find Sections [1](#), [5](#), [7](#), and [9](#) most directly relevant.

1.2 The Importance of User Research, UX/UI Design & Innovation

Digital technologies and digital transformation initiatives are rapidly shaping economic activities in Ghana. However, the variety of digital apps do not translate to widespread adoption especially within certain economic facets. While the reasons for low adoption differ across contexts, one common finding emerges: digital solutions are unlikely to achieve widespread uptake or deliver their intended impact unless they are designed to address users' needs, constraints, and the conditions that shape adoption. The potential of a solution is only realized when the tools actually work for the people they are supposed to serve.

User research, UX/UI design, and innovation are key elements in creating successful digital products and services. User research helps organizations understand users' needs, behaviors, and challenges, ensuring that products are designed to solve real problems rather than assumptions. By gathering feedback early in the development process, businesses can make informed decisions, reduce development costs, and improve customer satisfaction.

UX (User Experience) and UI (User Interface) design work together to create products that are both easy to use and visually appealing. UX focuses on making interactions simple, efficient, and enjoyable, while UI enhances the product through effective layouts, colors, typography, and interactive elements. Good UX/UI design improves usability, increases user engagement, strengthens brand identity, and encourages customer loyalty.

Innovation complements user research and UX/UI design by introducing creative ideas, new technologies, and improved processes that keep products relevant and competitive. Together, these three elements ensure that organizations develop solutions that meet user expectations while supporting business goals. By combining user insights, thoughtful design, and continuous innovation, businesses can deliver products that provide value, improve user experiences, and achieve long-term success.

As an open-source resource, this guide welcomes contributions from practitioners, researchers, designers, developers, and organizations working on inclusive digital products anywhere in the world, with particular relevance to sub-Saharan Africa and other contexts where similar user conditions apply.

1.3 About the Project and the Intermediary Cohort

The project (User Testing and Support to the Adaptation of Digital Solutions for Informal Entrepreneurs) was implemented between 2023 and 2026 across six regions of Ghana (Ashanti, Eastern, Northern, North-East, Upper East, and Upper West) through a two-phase implementation approach.

Phase One focused on supporting existing digital solution providers to adapt and improve their products through user research, usability testing, co-creation, UX/UI consultation, and field validation. Solution providers in Phase One financed their own product adaptations based on the research findings.

Phase Two focused on enabling early-stage innovators to design and develop entirely new digital solutions, based on the needs identified through user-centred design, venture-building training, user research, and an incubation programme, supported directly by technical coaching and software development expertise. This is the key structural difference between the two phases: Phase Two solution providers received hands-on technical and software engineering support to build their prototypes, where Phase One providers had to resource that work themselves.

The project ran across four structured work packages: capacity building, user research and usability testing, co-creation and adaptation support, and field validation. A parallel monitoring, documentation, and knowledge-sharing effort ran throughout all phases and culminated in this guide.

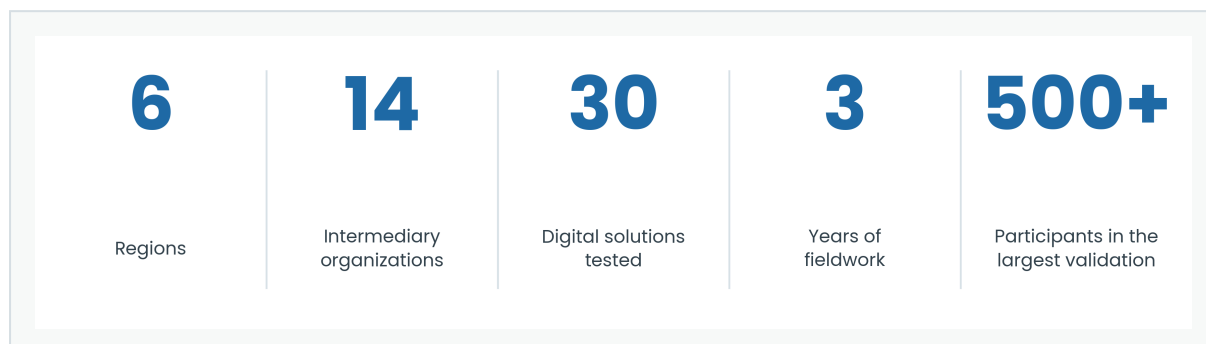


Figure 3 The project scope at a glance: 6 regions, 14 intermediary organizations, 30 digital solutions tested across both phases, 3 years of fieldwork

More than 59 organizations responded to the project's call for intermediaries. Twelve were shortlisted through a structured scoring process that assessed organizational capacity, community networks, gender inclusion orientation, logistics, and regional coverage. The twelve shortlisted organizations were brought through an intensive training programme in UI/UX design methodology, user testing, and gender-sensitive research before being assigned to specific solution partnerships.

How to read the reference codes. Intermediary codes reflect the order and batch in which organizations were recruited, not a strict region rule: most letter groups do correspond to one region (e.g. E1–E3 are all Eastern Region; A1–A5 are all Ashanti Region). Solutions are coded S1–S18 for Phase 1 (existing solutions adapted) and P1–P12 for Phase 2 (new solutions incubated). The same code refers to the same organization or solution every time it appears in this guide, so partnerships and case studies can still be followed end to end; the code just replaces the name. **The intermediary cohort and solutions tested (Phase 1: existing solutions, 12 intermediaries, 18 solutions)**

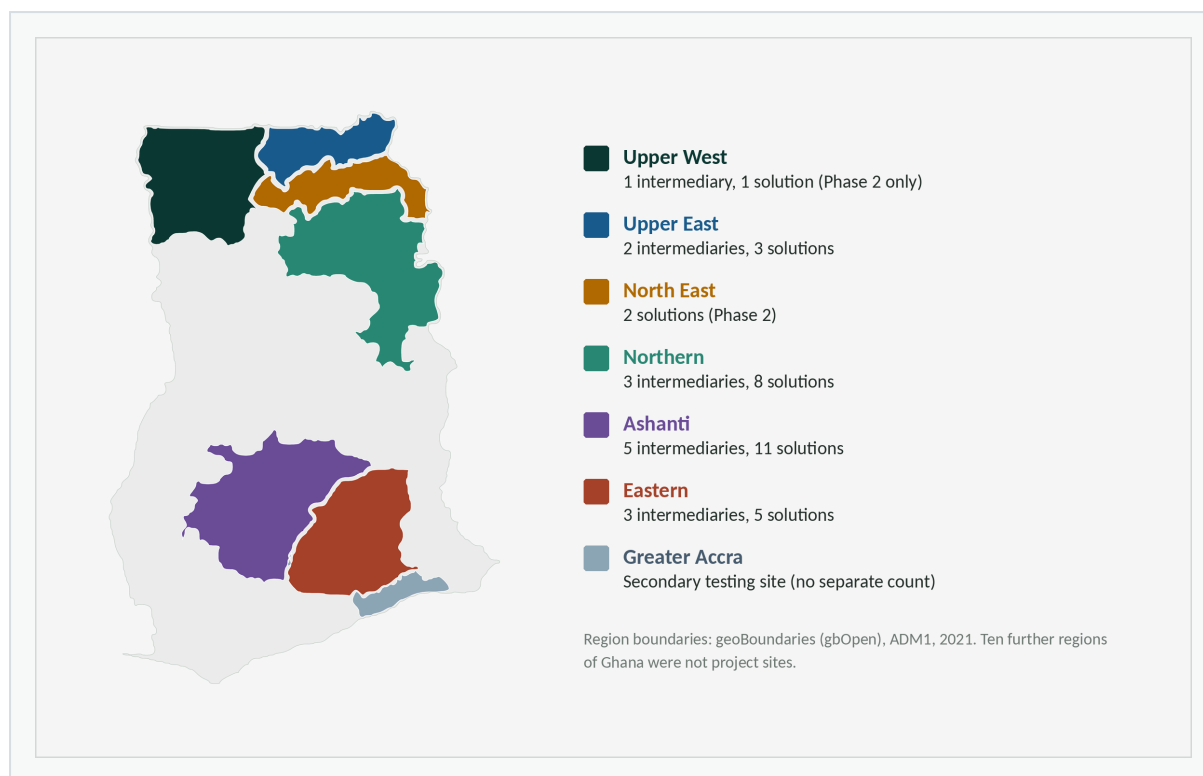


Figure 4 Where the project worked, by region

The intermediary cohort and solutions tested (Phase 1: existing solutions, 12 intermediaries, 18 solutions):

Upper East Region

- **Intermediary N1.** Tested Solution S1 with women traders, manufacturers, and business owners. Solution S1 was designed to unify order management, payments, and logistics across multiple channels.

Northern Region

- **Intermediary N2.** Tested two solutions: Solution S2 with women micro-enterprise owners (business management, invoicing, and payment tracking), and Solution S18, a remittance and payment app that was discontinued during the project.
- **Intermediary N3.** Tested Solution S3, a digital business management solution, covering financial records, customer management, and business operations.
- **Intermediary N4.** Tested two solutions: Solution S4, for business communication and customer outreach, and Solution S5, a loyalty rewards platform for women traders that was discontinued during the project.

Eastern Region

- **Intermediary E1.** Tested Solution S6, a digital business record-keeping and accounting solution, with women micro-entrepreneurs across the Eastern Region.

- **Intermediary E2.** Tested Solution S7, an inventory management system, with women entrepreneurs including differently-abled business owners in the Eastern Region.
- **Intermediary E3.** Led field validation of Solution S8 across Lower Manya, Asuogyaman, and Yilo Krobo districts, testing with women entrepreneurs in financial inclusion and credit access contexts. Sessions were conducted in Twi, Dangme, Ewe, and English.

Ashanti Region

- **Intermediary A1.** Led field validation of Solution S9, a crop e-commerce and cold store management solution, testing with women smallholder farmers and agricultural traders.
- **Intermediary A2.** Tested two solutions: Solution S10, a digital SME marketplace that was discontinued during the project, and Solution S11, a digital storefront and payment management application. Testing spanned both Greater Kumasi and Greater Accra, deliberately capturing differences between peri-urban and urban business contexts.
- **Intermediary A3.** Tested two solutions, both documented at roster level only (*see note below*): Solution S12, a financial management app, and Solution S13, an AI chatbot for business communications, Phase 1's one AI-enabled solution.
- **Intermediary A4.** Tested two solutions, both documented at roster level only (*see note below*): Solution S14, an integrated communications platform, and Solution S15, a business management and e-commerce visibility tool.
- **Intermediary A5.** Tested two solutions, both documented at roster level only (*see note below*): Solution S16, a business management tool (no further detail available in project records), and Solution S17, a data management, analytics, and reporting solution.

A note on the solutions tested. Several intermediaries facilitated the user testing of more than one solution, which is why Phase 1's solution count exceeds its intermediary count. Three of the eighteen were discontinued (Solution S5, Solution S10, and Solution S18) for reasons including provider capacity and shifting business priorities; the guide treats these as part of the honest record rather than omitting them. The remaining fifteen ranged from fully redesigned, launched applications (nine solutions) to redesigned prototypes still in development at the close of the project (six solutions). Solutions S12 through S18 (Intermediary A3, Intermediary A4, Intermediary A5's Phase 1 solutions, and Intermediary N2's Solution S18) are documented at roster level only in this edition; the field validation detail available for them is thinner than for the rest of the Phase 1 cohort, limited to the project's summary records rather than full validation reports.

Several solutions were still in active development at the time of testing. Solution S10, Solution S9 in some modules, and Solution S4 were among those where participants and intermediaries were providing feedback on an incomplete product. This required careful framing, honesty about the development stage, and a validation approach focused on design direction rather than live functionality.

Intermediary N2's partnership with Solution S2 raised a structural issue that recurred across the project: solution providers with limited financial and development capacity who genuinely could not commit the resources needed to act on co-creation feedback. This is a project design issue that future initiatives must address through provider readiness assessment before selection.

Phase 2: new solutions. Building on what Phase 1 surfaced, the project then developed a second track: an ideation-to-incubation programme that took new, early-stage digital solutions from concept through prototyping with the same target population, extending into two regions that Phase 1 did not reach (Upper West and North East) alongside Northern, Eastern, Ashanti, and Upper East. A call for ideas drew 176 applicants; 29 startups were selected into the ideation phase, and twelve high-potential ideas progressed into the incubation programme itself. Selected ideation teams went through structured rounds of design-thinking workshops, incubation with hub and technical mentors (including, unlike Phase 1, direct software engineering support to build their prototypes), iterative MVP testing with target users, and a final pitching event judged by a panel including academic, hub, and MSME representatives. Six intermediaries supported Phase 2: four continued from Phase 1 (Intermediary N1, Intermediary N3, Intermediary E2, and Intermediary A4), and two joined newly (Intermediary U1, Upper East, and Intermediary UW1, Upper West). In short: Upper West was reached for the first time in Phase 2, through Intermediary UW1. North East is harder to state with certainty, because the project's Phase 2 records list Intermediary N1 under North East while its Phase 1 records list the same organization under Upper East; this guide flags that inconsistency rather than silently resolving it (see the note on reference codes above). Upper East itself is not new to Phase 2 either way, since Intermediary N1 (under whichever region label) and Intermediary U1 both place Phase 2 activity there as well.

Of the twelve incubated solutions, six were selected as finalists at the closing pitch event and received in-kind prize support for further development. The twelve, with finalist status noted, are:

- **Solution P1** (Intermediary U1, Upper East Region: Zaare, Tamale, Navrongo, Bolga). An e-commerce marketplace connecting women artisans directly to buyers.
- **Solution P2** (Intermediary U1, Upper East Region: Builsa North Municipal). A booking platform connecting women smallholder livestock farmers with veterinary officers.
- **Solution P3** (Intermediary A4, Ashanti Region). *Finalist, 2nd place*. A voice-enabled bookkeeping application for informal-sector traders with low literacy.
- **Solution P4** (Intermediary A4, Ashanti Region: Asante Akim South District). A USSD-based platform connecting livestock farmers to veterinary services in under 30 seconds, distinct from Solution P2 despite the similar problem space.
- **Solution P5** (Intermediary N3, Northern Region: Tamale). A milestone-based savings and crowdfunding platform for women-led village savings and loans associations (VSLAs).

- **Solution P6** (Intermediary N3, Northern Region). *Finalist, 3rd place.* A digital savings-group (susu) management platform.
- **Solution P7** (Intermediary N3, Northern Region). A simplified, offline-first bookkeeping tool designed specifically for women entrepreneurs with little or no prior digital experience.
- **Solution P8** (Intermediary UW1, Upper West Region). *Finalist.* An on-demand platform connecting smallholder farmers to agricultural services.
- **Solution P9** (Intermediary N1, North East Region). *Finalist.* A real-time mobile money fraud-verification app.
- **Solution P10** (Intermediary E2, Eastern Region). An AI-enabled financial record-keeping tool with voice prompts for low-literacy users.
- **Solution P11** (Intermediary E2, Eastern Region). *Finalist, 1st place.* An embedded, data-driven, AI-supported platform combining crop insurance, micro-pension savings, and local-language advisory services for women smallholder farmers.
- **Solution P12** (Intermediary N1, North East Region). *Finalist.* A digital solution connecting farmers directly to markets to reduce post-harvest losses.

02

Understanding the Users: Informal Sector Entrepreneurs

An honest picture of the women whose businesses these tools have to fit.

2.1 The Ghanaian Context: Who the Users Are

Any meaningful work in this space starts with an honest picture of who the users are and what their daily lives look like. This section draws on participant profiles, field observations, and research findings gathered across all project partnerships.

Women micro-entrepreneurs in peri-urban and rural Ghana are not a homogeneous group. They range from young women running digitally powered, retail businesses in Tamale to older traders who have managed market stalls for decades using notebook records and informal networks. They include differently-abled entrepreneurs, smallholder farmers, food processors, hairdressers, tailors, fabric sellers, and agricultural traders. What they share is that most digital solutions were built without any of them in mind.

Critically, the goal of user research in this space is not to produce a “typical female user” profile. The aim is to develop multi-dimensional, intersectional personas that reflect age, geography, caregiving roles, literacy level, digital experience, language, and sector. A one-dimensional profile creates one-dimensional solutions. (Sample composite personas built from project participant data are provided in [Section 6.3.](#))

2.1.1 Socio-Economic Realities

The businesses involved in the project spanned retail, food processing, agricultural trading, crop e-commerce, inventory management, service provision, and manufacturing. Income streams were often irregular and cash-based. Many participants managed both business and household finances from the same pool of money, which made any financial tool with subscription fees or complex pricing a difficult proposition.

A consistent pattern emerged: a digital tool has to demonstrate visible value fast. If it does not make something easier or more profitable within the first few uses, it gets set aside. This is not resistance to technology; it is entirely rational behaviour from people who cannot afford to waste time or money on tools that do not deliver.

- Income is often seasonal, informal, or cash-based. Pay-as-you-go pricing is strongly preferred over subscriptions.
- Business and household finances frequently overlap. Tools that track only business transactions can feel incomplete or irrelevant.
- Mobile data costs are a genuine barrier. Solutions that consume significant data or require constant connectivity will be abandoned.
- Time has a direct cash value for traders. Any tool that slows down a transaction during peak hours will not survive daily use.
- Access to finance is a recurring challenge. Intermediary E2 found that when discussing digital tools with participants, women consistently cited access to finance as a major concern alongside tool usability.

Phase 2 testing put numbers behind how severe these constraints are. Among Solution P2's participants (women livestock farmers, Upper East Region), 93% had lost at least one animal to preventable disease within the past year, with reported losses of GHS 300–1,500 per animal: money that, in this context, is often earmarked for school fees or medical costs, not discretionary spending. Among Solution P5's participants (women-led MSMEs, Northern Region), all ten had been turned down by at least one formal lender in the past twelve months and were paying 40–60% interest on informal loans as a result. Solution P4's later validation in Ashanti Region added a frequency measure: its thirty farmer participants reported livestock health problems roughly once every two months, peaking in the rainy season. Figures like these are why pay-as-you-go pricing and low switching costs are not a nice-to-have for this user group; they are testing against a baseline of real, recurring financial precarity.

2.1.2 Cultural and Gender Dynamics

Trust networks are central to how women in these communities evaluate and adopt new tools. Both Intermediary N3 and Intermediary N1 found that women are far more likely to try something new when it is introduced by a peer, a market queen, or a trusted association leader than through a general advertisement or cold outreach. Solutions endorsed by familiar institutions consistently had higher uptake.

Caregiving and domestic responsibilities also shape adoption directly, not just as background context. Solution P2's testing surfaced this starkly in a comparison between two participants with the same underlying problem: *"The men can go to town and find help. But I have children to care for and cooking to do. I cannot leave my home for a whole day to chase a vet."*

A design or service model that assumes a user can freely travel or step away is, in practice, assuming away a constraint that falls disproportionately on women in this user group. Phone access itself can be gendered within a household. Solution P2’s team also found that some women farmers shared a phone with a husband or older child, which shaped who could act on a time-sensitive notification and when. Solution P4’s validation recorded the same dynamic from the other side: where older farmers struggled with the USSD menus, younger family members submitted requests on their behalf, an assisted-use pattern that service design for this population should expect and support.

Privacy matters. Shared household phones mean that some women cannot store business records on their personal device without those records being visible to family members. Solutions with PIN-protected sections, quick logout features, or separate profiles for business and personal use received noticeably more positive responses. Intermediary E1 specifically flagged shared device use as a design constraint that affects both privacy and the simplicity of onboarding flows.

Language is both a practical issue and a dignity issue. Using English as the sole language of a platform signals to many users that the product was not designed for them. Intermediary E2 found that even participants who spoke English preferred interacting in their local language during research sessions, because it enhanced engagement and produced more candid, detailed responses.

Intermediary N4’s experience in Northern Ghana pointed to the relevance of cultural commercial practices. The concept of *jara* (a small goodwill addition given to loyal customers beyond the formal transaction) is embedded in how women traders think about customer relationships. A cashback loyalty platform like Solution S5 aligned directly with this cultural practice, which influenced how participants evaluated its relevance and appeal.

FIELD OBSERVATION: INTERMEDIARY N1 + SOLUTION S1 (TAMALE, 2025)

During initial user testing, the password setup on Solution S1 was causing login failures before participants had even seen the app’s core features. Once simplified, a validation participant put it plainly: “This time I got in within less than a minute.” (The full co-creation cycle behind this fix, and why it became this guide’s model for writing evidence-backed recommendations, is in Section 4, Part A, Step 3.)

2.1.3 Digital Literacy and Business Acumen

The participant cohorts in the project showed a wide range of digital experience. Among Intermediary N1’s group in Tamale, 60 percent had basic or secondary school education, 30 percent had tertiary qualifications, and 10 percent had postgraduate qualifications. Digital literacy did not map neatly onto formal education: some participants with secondary schooling were confident mobile money users, while some with higher qualifications had very limited app experience.

Intermediary E2 made an observation that proved consistent across the project: literacy levels do not correlate with business acumen. Many participants with limited formal education demonstrated a sharp understanding of how to maximize profits, manage customer credit, and track stock. They were not digitally excluded because they lacked business intelligence; they were excluded because the tools were not designed for how they work.

Solution S11 testing by Intermediary A2 found that women micro-entrepreneurs increasingly have smartphones and can navigate through features when those features meet the needs of their business. The issue was not capability, it was relevance and accessibility.

A pattern emerged consistently across sites: even participants comfortable with mobile phones struggled when they encountered unfamiliar interface structures. Creating an account, navigating multi-step forms, and interpreting icon meanings were common pain points regardless of overall digital confidence.

FIELD OBSERVATION: INTERMEDIARY A1 + SOLUTION S9 (ASHANTI REGION, 2025)

Intermediary A1's validation of Solution S9 platform found that mobile-number-based login was successfully completed by all participants. However, 61.5% of participants needed assistance during initial sign-up, and the inventory update task had a 36.4% failure rate. Font and icon sizes were repeatedly flagged as too small, particularly for older participants. Five participants specifically requested voice features to guide them through key workflows. "Confused about the interface" and "Saw lots of options so I got confused" were the most common responses from participants encountering the platform for the first time.

Implications:

- Simplify onboarding. The first experience with a product should be completable without any external help.
- Use large, clearly labelled buttons. Small icons without text labels create confusion, particularly outdoors.
- Show clear confirmation messages. Participants frequently could not tell whether an action had succeeded or failed.
- Allow easy error recovery. Fear of making irreversible mistakes was one of the most consistent reasons participants avoided exploring a feature.
- Fix bugs promptly. Intermediary E1 noted that unresolved bugs undermine user trust rapidly, especially among first-time app users.

2.1.4 Environmental and Operational Context

Market stalls, roadside kiosks, home-based workshops, salons, hairdressing shops, and farm storage facilities are not comfortable environments for extended digital interaction. Screens are hard to read in direct sunlight. Background noise makes audio features unreliable.

Customers arriving interrupt any task, meaning tools must support quick pauses and easy resumption.

Most participants were multitasking constantly. A payment app requiring five taps to record a transaction is genuinely harder to use than a notebook when a customer is standing in front of you. The design bar for replacing informal methods is higher than many developers appreciate.

- Design for interruption. Tasks must be saveable and resumable without data loss.
- Minimise steps per common action. Recording a sale or checking a balance should take no more than two to three taps.
- Optimise for outdoor visibility. High-contrast text and large fonts are not optional features.
- Optimise for low-specification Android devices. Many participants used entry-level or second-hand smartphones. Solutions must perform on these devices.
- Test in realistic environments. Office or hub testing will not surface connectivity, lighting, or interruption issues that appear in actual market conditions.

2.1.5 Trust, Perceived Value, and Safety

Trust has to be earned before a solution can be adopted. Participants in the project were generally cautious about apps that asked for personal data, financial information, or access to contacts without explaining why. Intermediary N2 found that building trust through transparent communication about security, data privacy, and fraud protection was as important as the functional design of the solution.

FROM THE FIELD

“With the provision of available customer reps to help me, I no longer feel intimidated by the app in any way.”

Validation participant, Solution S1, Tamale, 2025.

Perceived value also operates on an emotional level. Women who felt that a solution had been designed with their needs in mind were more engaged and more forgiving of minor issues. Women who encountered interfaces that felt condescending or unnecessarily complex were much less likely to persist. Intermediary A2’s Solution S11 research found that women prioritize what is important to them in terms of app use and are eager to improve their digital capabilities when it clearly benefits their business.

03

Principles for Gender-Responsive UI/UX Design

The five commitments that gender-responsive design requires, and the evidence behind each.

These principles, drawn from the project’s combined field evidence and synthesized with inputs from the GIZ project team, represent the foundational commitments that gender-responsive digital design requires. They should inform product decisions at every stage, from the earliest concept to the latest iteration.

The principles do not replace specific design guidelines (Section 6 provides those). They are the “why” that gives the guidelines their direction. A designer who understands these principles can apply them even in situations this guide does not explicitly cover.

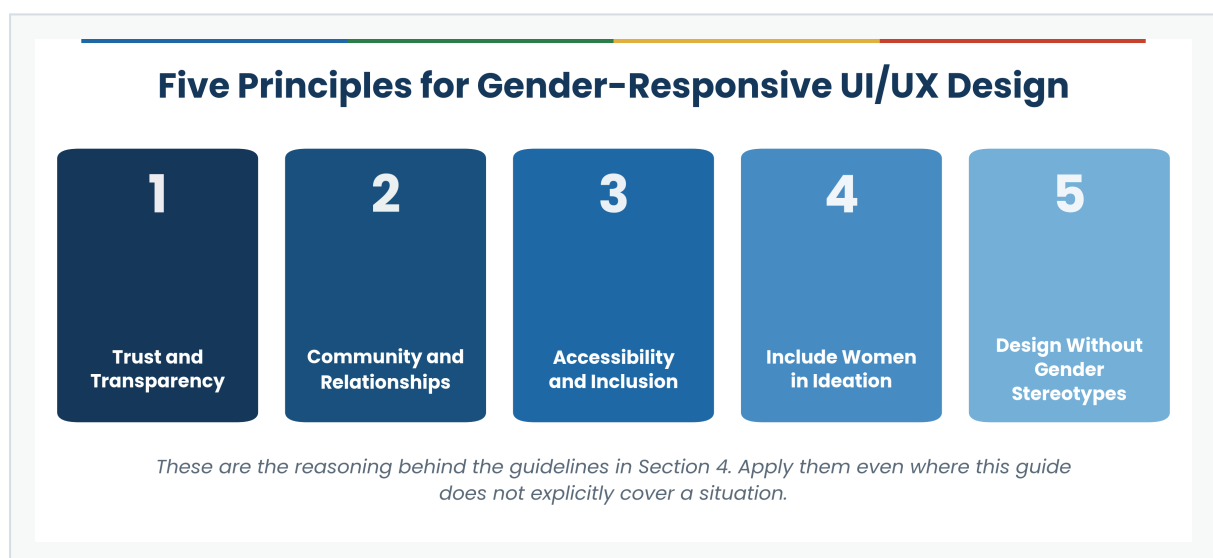


Figure 5 The five principles for gender-responsive UI/UX design

3.1 Trust and Transparency

Design for trust before asking for engagement.

Women in the project consistently made trust-based adoption decisions. A product that is unclear about its purpose, pricing, or data use will not be adopted regardless of its functional quality. In practice, this means:

- Use clear, plain language rather than technical jargon at every point of contact.
- Explain pricing and subscription terms directly, before sign-up, not buried in terms and conditions.
- Provide transparent explanations of privacy, data use, and fraud protection within the product itself, not only in documentation.
- Make human support visible. Knowing a real person can be reached increases willingness to try things independently.
- Show what happens after every action. Confirmation messages and status indicators reduce anxiety and prevent repeated actions.

Intermediary N2 and Intermediary A2 both documented that building trust through transparent communication about security features was as impactful as functional improvements. Solution S1's introduction of accessible customer support representatives had a measurable positive effect on user confidence even among participants who never needed to use the support line.

3.2 Community and Relationships

Design for the social context of use, not just the individual transaction.

Women micro-entrepreneurs in the project operated in deeply relational business environments. Their adoption of digital tools was shaped by peer recommendations, community endorsements, and shared experiences. In practice, this means:

- Build social proof mechanisms: testimonials, peer reviews, and visible adoption by trusted community members.
- Consider features that support community dimensions of business, such as group savings, peer referral, and shared order management.
- Support human-centered interactions. Automated responses alone are insufficient where trust is established through personal relationship.
- Enable sharing: the ability to share receipts, records, or reports with business partners, family members, or group leaders is a practical feature with high value.
- Design for peer learning. Products adopted through community networks are more sustainable than those dependent on individual digital confidence.

Intermediary N1's experience with Solution S1 showed that participants who adopted the platform with peer support were more consistent users than those who onboarded alone. Intermediary N4's Solution S5 work demonstrated how a loyalty platform that aligned with the cultural practice of *jara* gained immediate emotional resonance.

3.3 Accessibility and Inclusion

Design for the full range of users from the outset.

Accessibility in this context goes beyond international web standards. It encompasses language, literacy, connectivity, device capability, and physical ability, all of which vary significantly across the women who participated in the project. In practice, this means:

- Plain language as the default. Simple, concrete words that correspond to how users describe their own work.
- Local language as a functional priority, not a late-stage add-on.
- Voice assistance and audio guidance for key workflows, particularly relevant for users with lower literacy or visual constraints.
- High readability: adequate contrast, font sizes that work outdoors on small screens, and spacing that prevents mis-taps.
- Mobile-first design, since smartphones are the primary and often sole internet device for this user group.
- Designing for differently-abled users as a standard requirement, not a special case.

Intermediary E2's work with differently-abled women entrepreneurs demonstrated that the design improvements made for accessibility consistently benefited the entire user group. Larger text, better contrast, and simplified navigation helped older participants, users working in bright sunlight, and first-time smartphone users just as much as those with visual or physical constraints.

PRACTICAL TIP

Treat accessibility fixes as fixes for everyone, not a separate workstream. Every accessibility improvement documented in the project (larger text, higher contrast, simplified navigation, voice guidance) measurably helped users who had no stated accessibility need at all. Budget and design for accessibility in the first pass, not as a retrofit after testing with a “typical” user.

3.4 Include Women in Ideation, Not Just Testing

The most valuable gender-responsiveness work happens before the product is built, not as a retrofit. Intermediary E2 recommended conducting user testing and gathering feedback to understand the needs and pain points of women-MSME owners as a continuous process, not a one-off exercise. Intermediary N1 recommended participatory design workshops with women micro-entrepreneurs. Intermediary N4 recommended involving end users continuously, not just at the final stages of development.

3.5 Design Without Gender Stereotypes

Visual design choices carry meaning and shape first impressions. Avoid gender stereotypes in colour, typography, and imagery. Using pink or pastel palettes to signal that a product is “for women” is reductive and often counterproductive in professional business contexts. Colour, type, and imagery choices should be driven by legibility, trustworthiness, and context, not by gender signalling. Full visual design guidance is provided in Section [6.1.9](#).

04

UI/UX Design Best Practices When Designing for Informal Sector Entrepreneurs

How to run user testing step by step, for products that exist and products that do not yet.

The project was implemented in two phases with different starting points for each phase. **Part A** covers user testing and adaptation of **existing, already-built digital solutions** (Phase 1: the twelve intermediary partnerships described in Section [1.3](#)). **Part B** covers UI/UX practice for **new solutions built from scratch** through the project ideation-to-incubation programme (Phase 2). Both follow the same three-step logic (prepare, conduct, analyse and adapt), but the starting point, constraints, and pace differ.

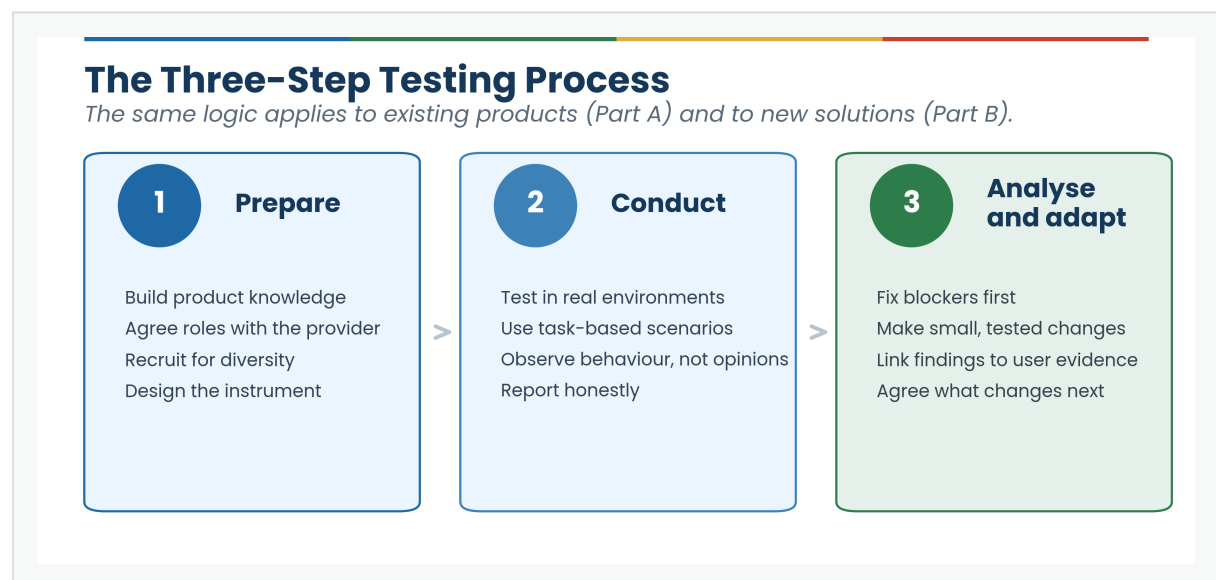


Figure 6 The three-step testing process, common to both parts of Section [4](#).

Part A: UI/UX Design for Existing Solutions (Phase 1)

Step 1: Preparing for User Testing

Train the intermediary team before fieldwork begins. The quality of insights from any user research effort depends heavily on the people doing the fieldwork. Organizations that arrived at fieldwork with a solid grounding in user testing methodology, gender-sensitive facilitation, and product knowledge produced far richer findings than those where preparation was rushed.

Training must address both technical and interpersonal dimensions. On the technical side, intermediaries need to understand what usability testing is actually for, how to run a task-based session, how to write neutral observation notes, and how to distinguish between a usability problem and a participant preference. On the interpersonal side, they need to be comfortable working with participants who are cautious, distracted, or uncertain about the process. Intermediary N1's input to this guide was specific: training must ensure intermediaries are adequately knowledgeable not just in user testing concepts but in the business applications of user interfaces relevant to the specific product they are testing. Intermediary N3 added that training must build confidence in iterative testing.

What training needs to cover:

- Cover user testing and user research concepts in full before moving to practice.
- Curate focused training materials, including checklists, flowcharts, and illustrated guides.
- Include training on gender and cultural sensitivity.
- Use role-play and practice scenarios; there is no substitute for practising facilitation before doing it for real.
- Train for diverse participant groups. Facilitation approaches must adapt to participants with very limited literacy, strong language preferences, or significant digital anxiety.
- Train intermediaries to document direct quotes and behavioural cues during sessions.

Understand the product before testing begins. Intermediary N1's recommendation was direct: product orientation sessions should be in person wherever possible, held at the solution provider's location, and should include hands-on time with the product rather than a slide presentation.

- Require the solution provider to deliver a full product demonstration before testing begins, using real business scenarios relevant to the target users.
- Use plain, non-technical language in product explanations.
- Give the intermediary team sandbox access to explore the product independently.
- Document known limitations. If a feature is still in development or has known bugs, the testing team must know this before interpreting participant behaviour.

Set up direct communication with the solution provider before testing starts, not after something goes wrong. In the project, communication gaps between the intermediary and the solution provider (about timelines, expectations, or who was responsible for delivering a testable version of the product) were a recurring source of delay, and in a future engagement without a programme manager coordinating between the two sides, an intermediary cannot rely on that layer to catch the problem. Agree upfront on a check-in cadence and escalation point with the solution provider directly. Intermediary N4's recommendation applies equally outside a managed project: escalate early when a solution provider goes quiet or fails to deliver a testable artefact, rather than absorbing the delay quietly and letting it compress the testing timeline. A regular check-in structure is not administrative overhead; it is how an intermediary protects the quality of its own findings.

Recruit and prepare participants. Effective recruitment starts with a clear definition of who you need: not just the target sector, but the range of digital experience, literacy levels, language backgrounds, age ranges, and business types that will give a complete picture of how the solution works across the intended user population. The most useful participant profiles in the project were deliberately diverse; over-relying on participants already comfortable with technology hides the barriers that prevent the majority of the target group from adopting a solution.

Community-embedded recruitment consistently outperformed generic outreach. Market queens, women's savings groups, business associations, and peer referral networks were the most effective channels across all three regions. Intermediary N3, Intermediary N1, and Intermediary A2 all reported that recruiting through established community structures produced participants who were more engaged, more representative, and more willing to give honest feedback.

- Start with your existing network, then expand through peer referral.
- Engage market queens and community leaders as recruitment partners, not just as access facilitators.
- Use trusted institutions: women's cooperatives, savings groups, churches, mosques, and market associations.
- Recruit more participants than the minimum requirement; Intermediary N1, Intermediary N4, and Intermediary A2 all recommended building an attrition buffer.
- Screen for the right mix. Intermediary E1 specifically recommended recruiting a mix of tech-savvy and less tech-savvy participants to capture the full usability picture.
- Prepare participants before sessions with a brief, jargon-free explanation of what usability testing is, why their opinions matter, and that there are no wrong answers. Intermediary N3 noted this reduced participant anxiety and increased the quality of verbal feedback.

- Uphold ethical responsibilities in recruitment: clear consent, anonymisation of all data, transparency about how findings will be used, respect for participants' right to withdraw at any point, and explicit consent before any recording.

Design research instruments for the participants, not the researcher. Research instruments should be structured around the participant's understanding of their own work, not around academic research conventions or reporting templates.

- Use open questions first, then structured follow-ups.
- Frame questions around the participant's business activities, not around app features.
- Avoid closed yes/no questions; Intermediary E1 recommended "show me how" or "walk me through" prompts.
- Avoid rating scales with more than three points for lower-literacy participants.
- Pilot every questionnaire with a small group before field use.
- Keep questionnaires to ten to fifteen minutes when combined with hands-on task testing (Intermediary A2). Organise questions from simple to complex, and close with an open prompt such as "What is the one change you would want to see in this app?"
- Combine structured and open methods. Intermediary A2 recommended moving beyond ratings and inviting participants to tell the story of what happened: "Tell us what happened when you tried to..." captures richer insights than "Rate this on a scale of one to five."
- Capture gender-specific dimensions explicitly: privacy from household members, freedom to use a phone during the day, comfort using a financial app in public, time constraints, and confidence levels before and after testing.

Step 2: Conducting User Testing

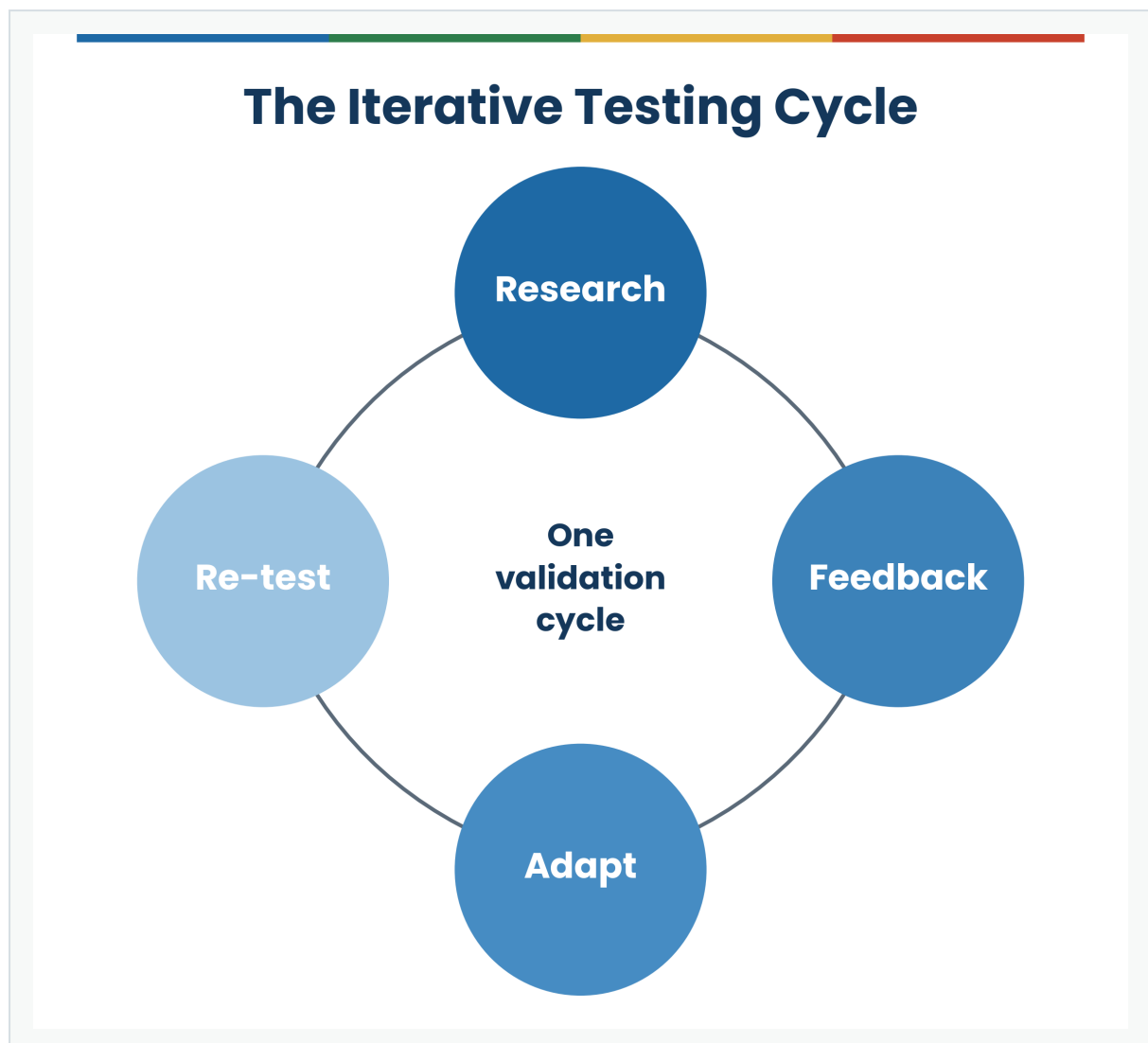


Figure 7 The project iterative testing cycle: research, feedback, adapt, re-test

The methodology used across the project was grounded in user-centered design principles, adapted for the specific realities of field research with women micro-entrepreneurs in peri-urban Ghana. Key methods applied across the cohort included interviews with women founders and business owners in local languages, contextual inquiry observing daily business routines in real environments, intersectional persona development, task-based usability testing with think-aloud protocols, and co-creation workshops where participants contributed directly to proposed adaptations.

Maintain the same participant cohort where possible. Returning users have prior experience with the product and are far better placed to evaluate whether changes have improved it. Intermediary N1 maintained the same cohort of ten women micro-entrepreneurs through to the final Solution S1 validation. This continuity produced particularly clear findings, since participants could point to specific things that had changed and assess whether those changes worked for them.

Run task-based sessions. Give participants specific tasks to complete on the product, observe while they complete them, and then ask structured questions.

- Design tasks that test the specific changes made during co-creation.
- Run one-on-one sessions wherever possible.
- Allow thirty to forty-five minutes per session.
- Allow users to try the product independently before guiding them.

Intermediary A1 applied this with Solution S9, structuring tasks around the features most used by agricultural traders: mobile login, stock entry, and inventory updates (the resulting task-completion data, including a 36.4% failure rate on inventory updates, is in Section [5.1](#)).

Test in real environments. Intermediary A2, Intermediary E1, and Intermediary N4 all emphasized that testing in participants' actual work environments (markets, shops, salons, kiosks, not just at hubs or training facilities) reveals barriers that controlled testing misses.

FROM THE FIELD: INTERMEDIARY N4 + SOLUTION S4 (NORTHERN REGION, 2025)

Intermediary N4's validation of Solution S4 platform included testing in market settings where participants typically conduct business. This context revealed that the time required to compose and send a promotional message was a genuine barrier during busy trading periods.

Participants asked for pre-written templates they could send with a single confirmation step.

Intermediary N4's recommendation: when a live app is unavailable during testing, use mockups or annotated printouts to ensure the session still yields directional findings.

Include both early adopters and slower adopters. Intermediary E1 made a point that is easy to overlook: early adopters tend to be more forgiving of rough edges; slower adopters reveal the barriers that prevent the majority of target users from getting to value. Balanced feedback needs both.

Combine observation and questioning. What participants do is often more informative than what they say. Facilitators observed hesitations, workarounds, re-reading of instructions, and points where participants stopped and asked for help: behavioural signals that did not appear in post-session questionnaires.

FROM THE FIELD

"I got through the login, but I do not think I could do it alone at home without someone to ask."

Participant during Intermediary N1 validation exercise, Tamale, 2025.

Document validation findings rigorously. Validation reports are evidence records, not marketing documents. State validation objectives clearly at the start. Report task completion rates for each task tested. Include direct participant quotes and visual evidence (before/after screenshots, session photos). Include all referenced appendices.

COMMON MISTAKES

Reporting only the positive findings. A validation report that omits the 36.4% task-failure rate because the 100% login-success rate looks better is not a validation report; it is marketing, and it will not help the solution provider fix what is actually broken. Report what happened, not what reflects best on the test.

Step 3: Analysing Results and Adapting UI/UX

Link every recommendation to user evidence. Every recommendation made to a solution provider should trace back to something a participant did or said. Vague recommendations like “improve the interface” rarely produce useful action; specific, evidence-backed recommendations do. The structure that proved most effective was: “*User feedback indicated [issue]. This led us to recommend [specific change].*” Intermediary N1 used this structure throughout their Solution S1 co-creation documentation.

CO-CREATION EXAMPLE: INTERMEDIARY N1 + SOLUTION S1

User feedback indicated: password creation caused errors and prevented women from completing registration. Recommendation: simplify the default password requirements. Change implemented: predetermined password system with simpler requirements. Validation result: every participant in the follow-up testing completed login without errors. “This time I got in within less than a minute.”

Prioritize what to adapt. Not everything can be changed at once, and solution providers need help prioritizing. The most important changes are those that block core usage: login failures, navigation dead-ends, error messages that cannot be dismissed. After those come changes that significantly reduce confidence or satisfaction. Aesthetic preferences come last.

- Identify blockers first: any issue preventing task completion must be resolved before anything else.
- Separate design preferences from usability problems.
- Be realistic about what a solution provider can change within the project timeframe.
- Document all findings, even those that cannot be acted on immediately.

Build a culture of small changes. One of the clearest lessons from the project is that small, well-targeted improvements consistently outperformed large redesigns. When a solution provider addressed multiple issues at once, quality suffered; when they focused on one or

two specific changes and tested them rigorously, the results held up in validation. Incremental improvement should start with whatever is preventing women from completing basic tasks; in this project, that meant **simplified logins, legible fonts, and offline functionality** appearing as early priorities across multiple partnerships. Intermediary E1 added a point that is often underweighted: fix bugs before anything else, since unresolved bugs undermine trust rapidly among first-time users.

FROM THE FIELD

“Even though I use reading glasses, I can read this clearly without squinting.”

Validation participant on the adapted Solution S1 typography, Tamale, 2025.

Provide pre-built, adaptable templates for common tasks rather than a blank canvas; Intermediary N4’s experience with Solution S4 showed women entrepreneurs are more likely to adopt a communication tool with ready-made templates they can customise. Test every change before moving on: run quick field trials with small groups, compare before-and-after performance (task completion, time on task, error counts), and observe as well as ask, since what participants say they found easy and what they actually found easy are sometimes different. Treat user confidence as a goal in itself, not just task completion: a participant who says “I think I did it right” has had a different experience from one who says “Yes, I know exactly what happened,” and confidence is closely tied to adoption. Document what changed and why for every co-creation cycle.

When providers are reluctant, slow, or unable to change. Resistance from solution providers happened in some partnerships. The most effective response was to return to the evidence: specific quotes, observation notes, and task completion rates. Intermediary N2’s partnership raised a structural challenge: solution providers with limited financial capacity who genuinely could not commit the development resources needed to act on co-creation feedback. This is a project design issue; provider readiness assessment before selection should include an honest evaluation of development capacity, not just intent. Intermediary N4 raised a related accountability concern: when solution providers fail to deliver testable artefacts on time, the intermediary’s work is blocked. Their recommendation was that intermediaries should escalate quickly rather than absorb delays quietly, and that even when a provider fails to deliver, the intermediary should document what would have been tested.

Part B: UI/UX Design for New Solutions (Phase 2)

Phase 2 tested a different kind of assumption. There was no existing product with real users to observe, only a problem hypothesis, and later a prototype, being validated for the first time.

In Phase 2 the promise was tested before the product fully existed.

Step 1: Preparing for User Testing (new solutions)

Test the problem before testing the product. Every Phase 2 team structured its validation around explicit, written assumptions in three tiers (problem assumptions, solution assumptions, and business-model assumptions), stated before fieldwork began and scored afterward as validated, partially validated, or not validated. This discipline is what Part A's Section 4.3 training curriculum produces for testing an existing product; for a new solution, the same discipline has to be applied one step earlier, to the problem itself. Solution P2's team, for example, went into the field with the explicit hypothesis that "a simple USSD-based coordination system can reduce delays in accessing veterinarians", a solution assumption they could confirm or reject rather than a vague intention to "test the app."

Recruit without an existing user base. Phase 1 intermediaries recruited through networks the solution provider or intermediary already had (Section 4, Part A, Step 1). Phase 2 teams by definition had no prior users, so recruitment leaned more heavily on the intermediary's own community standing: Solution P1's team worked through artisan associations in three Upper East towns; Solution P5 and Solution P6's team recruited through existing savings groups and market networks in Tamale; Solution P2's team worked through women's cooperatives in three named communities in Builsa North. Where a solution targeted a specific occupational identity (livestock farmers, artisans, savings-group members), recruiting through the trade or association structure that identity already had, rather than general public outreach, was consistently how teams found participants who could speak to the problem with authority.

Define literacy and digital-experience segments explicitly, not just demographics. Solution P7's team went further than most in operationalizing this: participants were classified as "Fully Digitally-Disconnected" (no digital skills, no awareness of what technology could do for their business), "Partially Digitally-Disconnected" (comfortable with digital tools socially but not for business), or "Digital Beginner" (uses business technology but struggles with complexity). This typology is worth adopting more broadly: it predicts what kind of support a participant will need far better than age or education level alone, echoing the Phase 1 finding that literacy does not correlate with business acumen (Section 2.1.3).

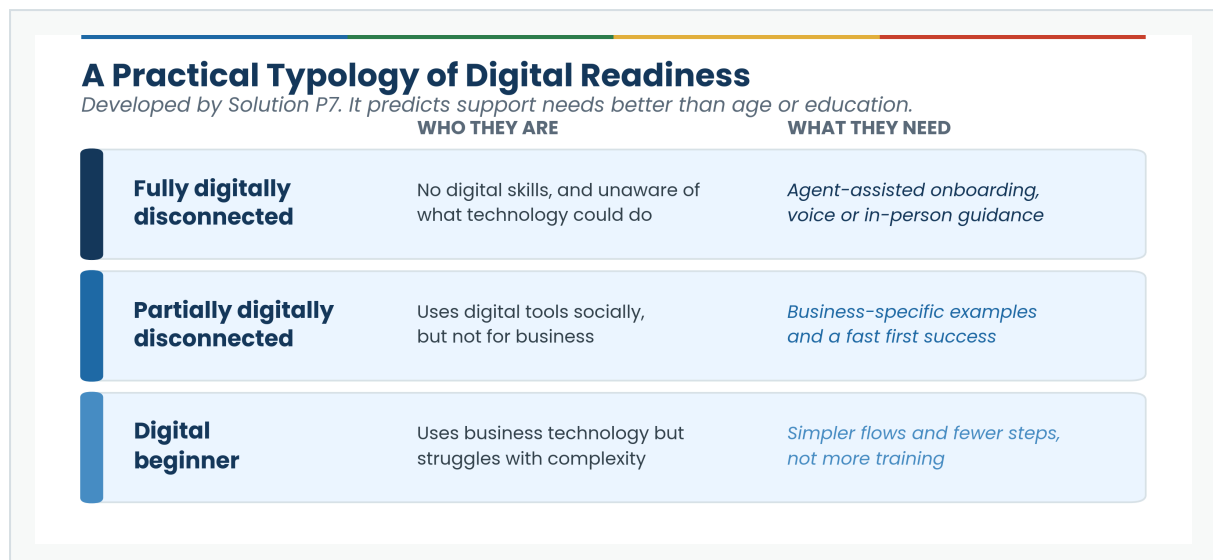


Figure 8 A practical typology of digital readiness, developed by Solution P7.

Step 2: Conducting User Testing (new solutions)

Match the testing method to what actually exists. Phase 2 teams tested at very different levels of product maturity, and adapted method accordingly: Solution P5 ran guided task-based sessions on a working iOS app distributed via TestFlight; Solution P2 combined a live website demonstration with a structured role-play, where a team member played the part of a dispatched veterinarian and physically arrived at the agreed time so participants could experience what the promise would feel like; Solution P1 combined in-person and phone structured interviews with live MVP demonstration for participants who had smartphones, and demonstration-plus-verbal-feedback for a participant who did not own one. None of these teams waited for a finished product before testing. The role-play approach in particular is a low-cost way to validate a service-delivery promise (not just an interface) before any code is written.

Watch for the gap between stated interest and structural exclusion. Solution P1's testing surfaced a participant who was highly motivated and articulate about the problem but could not use the platform at all because she had no smartphone access, a finding the team correctly treated not as a disqualifying edge case but as a structural exclusion issue requiring its own solution (an agent-assisted onboarding model), consistent with the Phase 1 principle of accessibility as a standard requirement rather than a special case (Section 3.3).

Let participants react to a concrete promise, not an abstract concept. The strongest signal across Phase 2 testing came from making the value proposition specific and checkable. When Solution P2 showed a booking confirmation with a specific arrival time, one farmer's reaction was immediate and telling: *"It tells me exactly when he will come? And he WILL come?"* Vague descriptions of "a platform that connects you to vets" would not have produced that reaction or the skepticism that followed it; both are more useful than polite general approval.

Step 3: Analysing Results and Adapting UI/UX (new solutions)

Distinguish “the problem is real” from “this solution addresses it.” Every Phase 2 validation report scored these separately, and the two did not always move together. Solution P5’s problem and solution assumptions were both strongly validated. Solution P4’s problem assumption was strongly validated (farmers have normalized unsafe self-medication because they have given up on formal veterinary access) but the solution only partially addressed it: the underlying constraint turned out to be too few participating veterinarians, not the coordination mechanism itself. Treating these as separate questions, the way Phase 1’s evidence-linkage discipline treats specific usability findings separately from overall product viability (Section 4, Part A, Step 3), keeps a team from either abandoning a valid problem because the first solution attempt was imperfect, or persisting with a solution the evidence does not support.

Take feature requests that exceed the MVP’s scope seriously; they are evidence, not scope creep. Solution P7 was deliberately built as a minimal three-action tool (record a sale, record an expense, view a summary), and it worked: participants completed the core flow and understood its value. But the same participants immediately asked for invoicing, point-of-sale, and receipt generation: features well beyond the MVP. Solution P5’s beta users, similarly, asked for credit scoring and savings tracking beyond the crowdfunding feature they were testing. Read together, these are not two teams that under-scoped their MVPs; they are a consistent signal that once this user group trusts a simple tool enough to adopt it, they want it to grow into a fuller business platform. That is a roadmap finding, not a reason to have built more before testing. Solution P7’s team was correct to launch minimal first and let demand for complexity surface through evidence rather than assumption.

Price against the specific alternative, not in the abstract. Willingness-to-pay data across Phase 2 was strongest where teams anchored pricing to the cost of what participants already tolerated. Solution P5’s 5% platform fee tested well specifically because participants compared it to the 40–60% rates charged by informal lenders they currently used, not to an abstract notion of “a reasonable fee.” Solution P2’s farmers, by contrast, resisted any upfront payment regardless of amount, because the specific alternative they feared was paying and then having no vet arrive; the objection was about payment sequencing and trust, not price level, which is why shifting to pay-after-service (and, for Solution P4’s veterinarians, a platform-held escrow model) addressed it where a lower price would not have.

Solution P11’s validation added two further pricing findings at larger scale. First, bundling raised willingness to contribute: insurance, micro-pension savings, advisory services, and post-loss recovery support offered through one trusted platform read as more valuable to participants than any of those products alone, and farmers valued recovery support after a loss almost as highly as the insurance payout itself, a reminder that what this user group is buying is resilience, not a policy. Second, contribution flexibility mattered as much as the amount: schedules matched to seasonal farm incomes, paid in small instalments over USSD, made commitment possible for participants for whom a fixed monthly fee would not have been.

Expect trust, not features, to be the binding constraint for service-coordination solutions. Solution P2 and Solution P4 both connect farmers to a human service provider (a veterinarian) rather than digitizing a task the user already does alone, and both encountered the same core barrier: years of unreliable informal alternatives had taught participants to expect no-shows, so the product had to overcome scepticism before its features mattered at all. Solution P4’s assessment stated this directly: *“trust and proof of effectiveness will determine adoption”*. Both teams’ next steps centred on visible, bounded pilots (a defined response-time target, a small committed cohort) designed to generate the first success stories rather than on adding product features. This is a distinct pattern from Phase 1, where the binding constraint was more often a specific interface friction (Section [5.3](#)); for a new two-sided service marketplace, the constraint is more often reputational, and no interface change fixes it faster than a demonstrated track record does.

FROM THE FIELD

“When animals fall sick, we don’t call anyone, because we know the vet won’t come. By the time help comes, the animal is already dead.”

Farmer participants, Solution P4 validation, Asante Akim South District, 2026.

Solution P4’s follow-up validation, run in January 2026 with thirty farmers and five veterinary professionals in Asante Akim South District, added a supply-side dimension to this pattern. The three-step USSD request flow itself worked: registration needed hands-on assistance, but most farmers could then submit requests independently, and they valued reaching several trained professionals at once instead of hunting for individual phone numbers. The main interface complaint was simply that the USSD shortcode was too long to remember easily. The real bottlenecks sat on the service side: the veterinarian pool was small, the system allowed a vet to accept only one request at a time, and vets showed a preference for giving remote advice over travelling to farms, all of which undermined the response-time reliability that farmer trust depends on. The revenue insight ran in the same direction. Vets were hesitant to pay for access to farmers, since demand already exceeded their capacity, but showed strong interest in paying for data analytics, case summaries, and periodic activity reports. For a two-sided service marketplace, the side that needs the platform most is not automatically the side that will pay for it, and the stronger paying proposition may be data, not leads. The team’s planned response follows the bounded-pilot pattern above: a structured three-month pilot with defined service-level targets, aggregation of requests by community so that one vet trip serves several farmers, emergency prioritization, and a response-time target under 24 hours.

05

Insights and Learnings

What the field work actually showed, including what did not work.

Three years of fieldwork across the full project cohort (the six regions, fourteen intermediaries, and thirty solutions introduced in Section [1.3](#)) produced a consistent message: the gap between a digital product and the women it is supposed to serve is almost never a technology gap. It is a design gap, a communication gap, and sometimes a commitment gap.

The products that worked better at the end of the project than at the start did not become different products; they became more specific to their users. Simpler logins, bigger text, support in local languages, pre-written templates, offline capabilities, and a phone number to call when something goes wrong. These are not complex innovations. They are the basics of designing for a real person in a real context.

What is harder than the design changes is the process that produces them: genuinely listening to users, building enough trust that participants will tell you when something does not work, maintaining that feedback loop over time, and having solution providers who are willing to act on what they hear. The Intermediary N2 and Intermediary N4 partnerships documented what happens when provider capacity or commitment is insufficient. The Intermediary N1 and Intermediary A2 partnerships documented what is possible when the collaboration is genuine.

5.1 Digital Literacy and Adoption Patterns

Digital literacy across the project's participant pool was more heterogeneous than initial assumptions suggested. Intermediary N1's cohort in Tamale showed 60 percent with basic or secondary education, 30 percent tertiary, and 10 percent postgraduate qualifications. These educational categories did not predict digital confidence reliably.

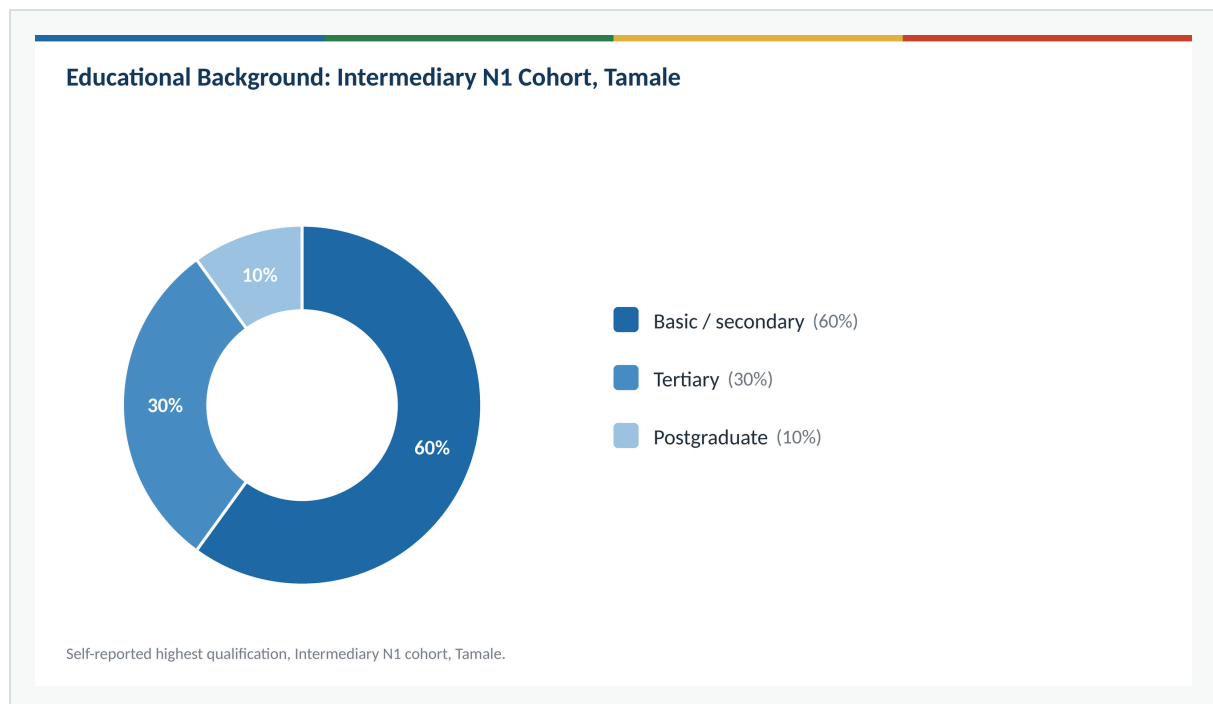


Figure 9 Educational background of the Intermediary N1 cohort, Tamale

Intermediary A1's Solution S9 data provided some of the most precise field metrics in the project:

- 100% of participants successfully completed mobile-number-based login, confirming that this is the right authentication approach for this population.
- 61.5% needed assistance during initial sign-up, indicating that standard onboarding flows are not yet calibrated for this user group.
- 36.4% failed the inventory update task, directly attributable to interface complexity and small tap targets.
- 5 participants independently requested voice guidance features, representing a consistent and unprompted signal about a significant accessibility gap.

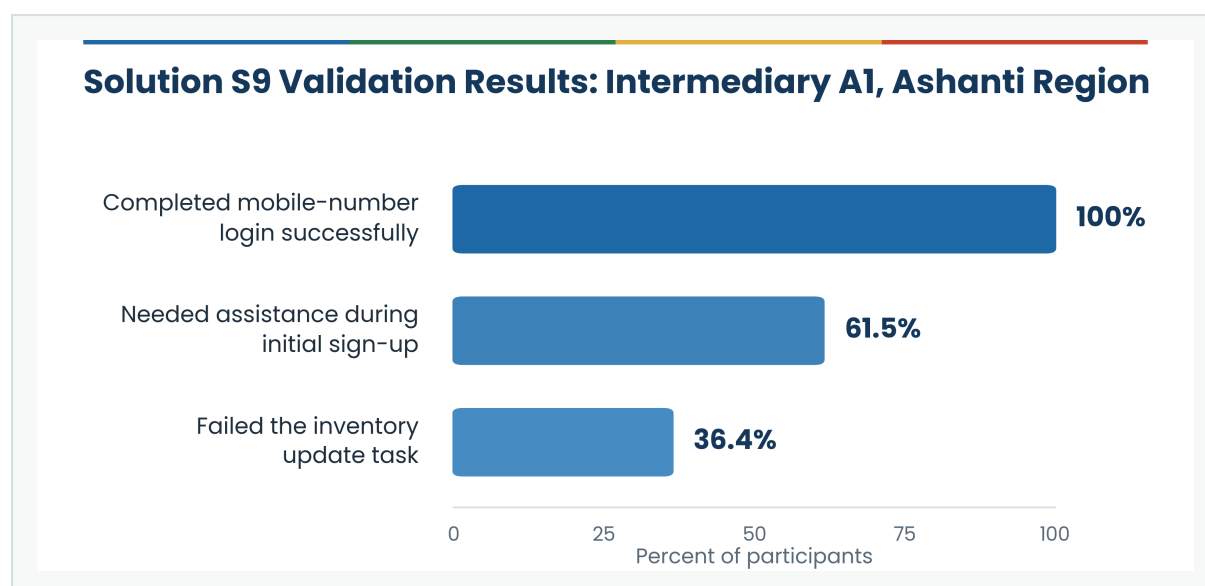


Figure 10 Solution S9 validation results, Intermediary A1

5.2 Language and Localization Evidence

The language data from this project is unambiguous. In every region, participants engaged more deeply, gave more honest feedback, and completed tasks more successfully when sessions were conducted in their first language. Intermediary E3’s validation sessions in the Eastern Region used four languages: Twi, Dangme, Ewe, and English. The multilingual approach was not a logistical complication; it was what made the findings representative. Single-language testing in English would have systematically excluded the participants most likely to represent the barriers the project was trying to address.

5.3 Task Completion and Friction Points

Across all partnerships, the tasks with the highest failure rates shared a common characteristic: they involved either account creation, multi-step navigation, or interface elements too small for reliable use on older devices. These are not edge cases; they are the core onboarding and daily-use flows that determine whether a product is adopted at all.

The most consistently successful tasks were those involving a single action with an immediate visible outcome: recording a payment, sending a message, or viewing a product price. These one-step flows had near-universal completion rates across literacy and digital experience levels. This is the design benchmark all core features should aspire to.

5.4 Co-Creation Impact: Before and After

Where before-and-after data was available, the impact of co-creation adaptations was measurable and consistent. The Intermediary N1/Solution S1 co-creation cycle produced the most documented improvement in the project: a password simplification that moved login from a common point of failure to a near-universal success in the validation phase.

The pattern repeated across other partnerships. Where specific, evidence-backed changes were implemented before validation, task completion rates improved and user confidence scores increased. Where the co-creation phase was compressed or where providers were unable to implement changes, validation data showed little improvement from the baseline. This data makes a clear argument for adequate time and genuine provider commitment in co-creation phases: it is the portion of the project cycle that converts research into measurable impact.

5.5 Key Behavioural Observations

Across all field sites, facilitators documented consistent behavioural patterns that structured survey data alone would not have captured:

- Participants who encountered their first unfamiliar interface element often stopped and looked to the facilitator before attempting to proceed, even when they had been told they could try freely. This confirms the need for confidence-building in onboarding.
- When confirmation messages appeared after a successful action, participants visibly relaxed. The absence of confirmation caused repeated actions and anxiety about whether tasks had been completed.
- Participants consistently found ways to describe a digital tool's value in terms of their existing business practices. A tool was “like my notebook but faster” or “like mobile money but for stock.” Designers should build on, not replace, these mental models.
- Women who participated in group sessions were more likely to compare experiences and point out usability differences than those in one-on-one settings, making group sessions particularly valuable for surface-level issue identification.

5.6 Phase 1 and Phase 2: What Carried Over, What Differed

Testing an existing product (Phase 1) and testing a brand-new one (Phase 2, Section [4](#) Part B) are different exercises, but the project evidence shows the same underlying principles governed both.

What carried over. Trust was the binding constraint in both phases, not technical sophistication. Phase 1's Intermediary N1/Solution S1 password fix and Phase 2's Solution P2 booking-confirmation reaction are, structurally, the same finding: a small, concrete, verifiable promise did more for adoption than any amount of feature depth. Local language and voice interaction moved from “accessibility feature” to “core requirement” in both phases; Phase 1's language findings (Section [5.2](#)) and Phase 2's Solution P3, where voice input started as a secondary feature and became the most-valued one, reached the same conclusion independently. And in both phases, the most reliable recruitment channel was an existing community structure the intermediary or team already had standing in, not general outreach. Solution P11's validation later confirmed this at the largest scale in the project:

more than five hundred women farmers reached through Farmer-Based Organizations and community leaders, with community trust proving a stronger driver of adoption than the technology itself.

What differed. Phase 2 teams could test a specific, falsifiable promise before a product fully existed (a role-play, a demo, a paper flow) in a way Phase 1's adaptation work, constrained by whatever the existing product already was, could not. This let Phase 2 validate or reject a business model earlier and more cheaply. Phase 2 also surfaced a pattern Phase 1's evidence base did not clearly show: users who adopt a deliberately minimal tool quickly ask for more capability once they trust it (Solution P7, Solution P5): a “start minimal, then earn the right to add complexity” dynamic that is a useful complement to Phase 1's “small changes, tested rigorously” discipline (Section 4, Part A, Step 3), not a contradiction of it.

5.7 Limitations and Ethical Considerations

This guide is grounded in three years of field activity, but it has boundaries. Understanding those boundaries is essential for applying the guidance responsibly.

Scope limitations. The findings in this guide reflect a specific project context: fourteen intermediary organizations, thirty digital solutions across Phase 1 and Phase 2, six regions of Ghana, and a participant population of women micro-entrepreneurs in peri-urban and rural settings between 2023 and 2026.

- Context-specificity. Findings from Tamale do not automatically transfer to Kumasi, and findings from Ghana do not automatically transfer to other countries. Practitioners applying this guidance elsewhere must conduct their own baseline research.
- Solution mix. The solutions tested were weighted toward business management, payments, and agricultural e-commerce. Sectors not represented, including health, education, and creative industries, may present different design challenges.
- Participant selection. Participants were recruited through community networks and intermediary relationships. This may have produced cohorts that were more digitally engaged than the full population of women micro-entrepreneurs in each area.
- Project timeline. Some solutions were still in development during testing. Findings about incomplete products reflect a snapshot in time and should be interpreted as directional rather than definitive.
- Depth of evidence varies by solution. Solutions S1–S11, P1–P7, and P11 have full field validation reports behind them (quotes, task-completion data, iteration history). Solutions S12–S18, P8–P10, and P12 (eleven solutions in total, added in this edition) are documented at roster level only (name, region, status, brief description), since no validation-report-level detail was available in the project's records at the time of writing. Treat roster-level entries as confirmed to exist, not as evidence of specific usability findings.

Ethical responsibilities. Practitioners using this guide to design or research digital solutions have ethical responsibilities that go beyond the technical guidance provided here:

- **Informed consent.** All research participants must understand what they are participating in, how their data will be used, and their right to withdraw. This is not a procedural formality; it is the foundation of research that respects participants' autonomy.
- **Data privacy.** Research data collected from women participants must be stored securely, anonymised in publications, and not shared with third parties without explicit consent. This applies to quotes, images, and demographic information.
- **Power dynamics.** Researchers from outside a community hold a degree of power over participants that can distort findings if not actively managed. Facilitation approaches should equalize, not amplify, power differences.
- **Representation.** Findings should not be generalized beyond the evidence. Saying "women in Northern Ghana prefer X" based on a cohort of fifteen participants in Tamale overstates both the evidence and the homogeneity of the population.
- **Do no harm.** Research activities and product deployments should not expose participants to financial, reputational, or privacy risk. Where a solution has known limitations, participants must be informed before use.

Designing with context awareness. The most important ethical principle in this space is humility: recognizing that designing for users whose context differs from your own requires active effort to understand that context rather than projecting assumptions onto it. This guide provides evidence from the field. It does not remove the responsibility to go to the field yourself, listen carefully, and adapt. The women who participated in the project gave their time and their honesty precisely because they wanted better tools. Practitioners who use this guide carry a responsibility to honour that investment.

5.8 AI in This Project: Evidence and Open Questions

Artificial Intelligence (AI) appears in this project's record in a small but growing set of solutions. In Phase 1, Solution S13, an AI chatbot for business communications, is documented at roster level only. In Phase 2, five solutions carry an AI component: Solution P3's bookkeeping application is built around an AI speech interface; Solution P10 combines financial record-keeping with voice prompts for low-literacy users; Solution P11, the closing pitch event's first-placed finalist, applies AI-powered satellite monitoring to crop and climate risk assessment in its agricultural insurance platform; Solution P9 uses AI in mobile money fraud verification; and Solution P8 uses AI for agricultural service matching. Of these, Solutions P3 and P11 carry full field validation evidence in this edition, and their findings point in usefully different directions.

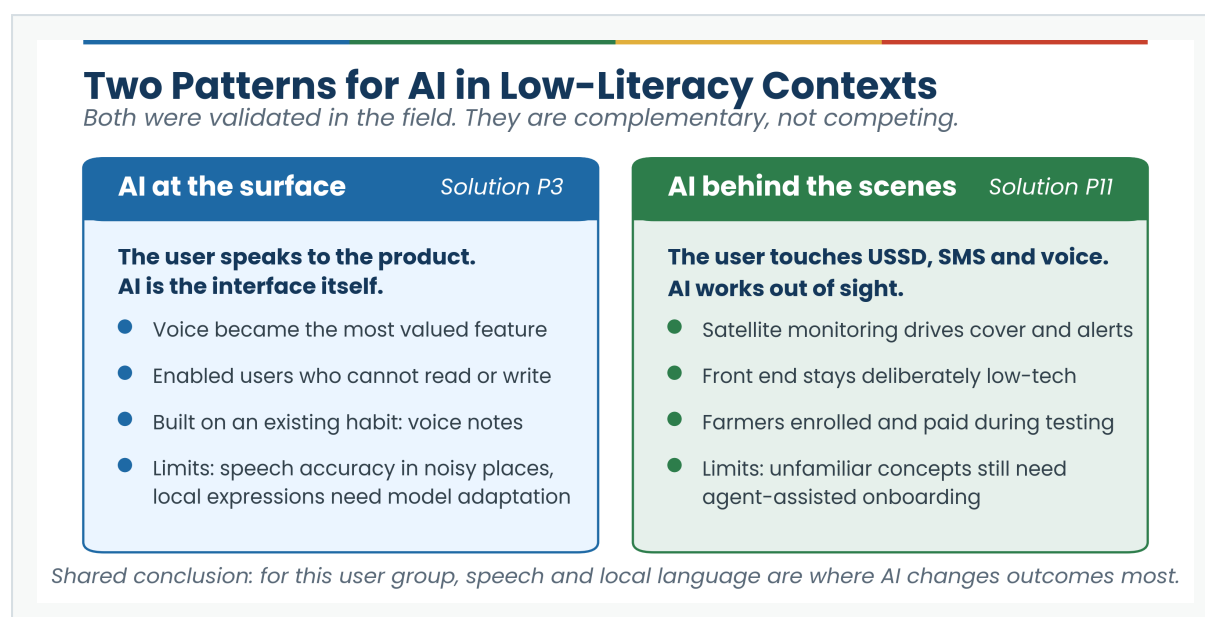


Figure 11 Two validated patterns for applying AI in low-literacy contexts.

Solution P3's validation is the project's clearest evidence that AI can remove barriers for this user group rather than add them. Voice interaction was not the MVP's original primary focus, yet it became the most valued feature in testing. Speaking to the application reduced friction, increased confidence during use, and enabled participation by users who could not read or write; participants also connected it naturally to an existing habit, sending voice notes on WhatsApp. Where voice guidance was available, participants navigated successfully; where it was limited, they got stuck. Participants asked for deeper conversational interaction, such as hearing account balances and transaction summaries spoken back to them, and approximately 80% of the seven participants indicated willingness to pay if the tool demonstrably improved their business organization, although payments were not enforced during testing, so that commercial signal remains unproven. The team's resulting decision was to raise voice-based workflows to roughly 90% of user interactions.

The same validation exercise documents the AI-specific work this requires. Speech recognition accuracy dropped when audio clarity was low, a real constraint in the markets and workshops where this user group works. Local business expressions and informal ways of stating amounts required adaptation of the underlying model; an off-the-shelf speech model does not arrive understanding how a Ghanaian trader phrases a credit sale. And connectivity still affected some features despite partial offline capability, which matters because speech processing tends to depend on a connection precisely for the users that offline functionality is meant to protect.

Solution P11 demonstrates a second, complementary pattern: AI in the back end, simplicity at the front. Its validation, run in early 2026 with more than five hundred women smallholder farmers through community sensitization meetings, product demonstrations, interviews, and focus group discussions, tested a platform whose AI works invisibly, in satellite-based monitoring of crop and climate risk that drives insurance cover and early-warning

advisories. What farmers actually touch is deliberately low-tech: USSD enrolment, SMS alerts, automated voice advisories in local languages, and field agents for assisted onboarding. Among the features drawing the strongest response were the local-language voice advisories and climate early warnings, and participants asked for advisory coverage in more local languages than the platform currently supported, the same direction of demand documented for Solution P3. During the validation period itself, farmers began enrolling and making small contributions through the USSD channel: early evidence that the value AI generates reaches this user group most reliably through the simplest available channels.

The demand signal extends well beyond one solution. Five of Intermediary AI's participants independently requested voice guidance during Solution S9 testing (Section [5.1](#)). Older farmers in Solution P4's validation struggled with USSD menus and asked for a voice version of the service. And local language moved from accessibility feature to core requirement in both phases (Sections [5.2](#) and [5.6](#)). Read together, these findings point to one conclusion: **for low-literacy users, speech interaction in local languages is the most consequential application of AI this project observed**, well ahead of more visible uses such as chatbots.

None of this exempts an AI feature from the tests this user group applies to every tool: visible value within the first few uses, reliable behaviour on low-end devices and unstable connectivity, trustworthy handling of money and data, and availability in the user's language (Sections [2](#) and [3](#)). A guided voice flow that reliably records a sale is worth more than an open-ended assistant that sometimes misunderstands one. Field evidence on AI-enabled solutions, including validation data for Solutions S13, P8, P9, and P10 as they mature, is a priority contribution area for future editions of this guide (Section [9.1](#)).

06

Open-Source Toolkit and Resources

The framework, patterns, personas, checklist and templates, ready to use.

This section is the practical reference: the design framework, functional patterns, checklists, and templates that put Sections [2–5](#) into practice. Use it as a standing reference at every stage of a project, not only when the guide’s narrative order reaches it.

PRACTICAL TIP

Short on time? Start with the Checklist (6.4), not the full Framework (6.1). The Checklist compresses this whole section into yes/no prompts you can run through in minutes. Come back to 6.1’s framework and 6.2’s patterns when a checklist item needs more context than a single line can give.

6.1 UX/UI Design Framework

6.1.1 Language and Literacy

Products that operate only in English in communities where Twi, Dagbani, Dagban, Gonja, Mampruli, Dangme, Ga, or Ewe is the working language are inaccessible by design (see Section [3.3](#) for the underlying evidence). Intermediary A2 recommended Twi and English bilingual options as a minimum for solutions deployed in the Ashanti and Greater Accra areas. Intermediary N4 specifically cited Dagban and Gonja for solutions targeting Northern communities. Intermediary N1 cited Dagbani, Mampruli, and Gonja for the Tamale ecosystem. Intermediary A2’s Solution S11 recommendations went further, naming Twi, Ga, Ewe, Hausa, and English as the full range of languages solutions operating at national scale would need to consider.

- Prioritize local language for onboarding flows, payment confirmations, and error messages.

- Use plain, concrete language throughout. “Today’s Sales” is clearer than “Revenue Summary.” “Stock Remaining” is clearer than “Inventory Balance.”
- Test comprehension of icons and symbols explicitly. Do not assume standard interface icons carry the same meaning across cultural contexts.
- Provide audio support for key instructions where literacy levels are low.
- Work with local language experts on translations, not just automated tools.

6.1.2 Connectivity and Offline Functionality

Every partnership encountered connectivity as a usability barrier. The most workable model identified across the project was local storage with background synchronization: the app stores data locally and syncs when connectivity is restored, without requiring any user action.

- Core functions must work without an active internet connection.
- Synchronize automatically when connectivity is restored.
- Show connection status clearly within the app.
- Minimize data requirements. Optimize images and load sizes for slow connections.
- For some contexts, provide USSD access as an alternative to app-only access.

6.1.3 Privacy and Safety

Shared-device use is a documented constraint for this user group (Section [2.1.2](#)); translate it into product requirements directly:

- Include a quick-logout or session lock option.
- Allow PIN-protected access to financial data.
- Provide a plain-language privacy explanation during onboarding.
- Build clear communication about security features, data privacy measures, and fraud protection into the product itself.

6.1.4 Onboarding and Support

Building on Section [2.1.3](#)’s onboarding findings: multiple intermediaries recommended short explainer videos in local languages as the most effective onboarding support tool for this user group. Intermediary E1 recommended a clear animated introduction to the solution before presenting features, breaking down functionality with step-by-step demonstrations using real-life examples, and adding tooltips and contextual help for every major feature.

FROM THE FIELD

“Just knowing that someone is there to call if I get stuck makes me more willing to try things on my own.”

Validation participant, Tamale, 2025.

Human customer support, as introduced by Solution S1, was seen as reassuring even by participants who never needed to use it. The mere presence of a support option changed willingness to explore the platform.

6.1.5 Time-Sensitivity and Task Efficiency

Time pressure on this user group is well established by this point (Sections [1.2](#), [2.1.4](#)); the practical translation is to make speed a design constraint, not an afterthought. Intermediary A2’s observations confirmed that features which save time are adopted faster than features that are functionally rich but slower to use.

- Common actions must be completable in two to three steps.
- Support partial task completion. If a user is interrupted mid-process, their progress should be saved.
- Provide automated receipts and quick-entry sales logs to reduce transaction recording time.
- Avoid unnecessary confirmation screens for non-destructive actions.

6.1.6 Designing for Differently-Abled Users

See Section [3.3](#) for the evidence behind this. In practice: budget for text size, colour contrast, touch target size, audio guidance, and navigation simplicity from the outset, not as a retrofit. These requirements go beyond standard adjustments for low-literacy users, and the target population for gender-responsive design is not a single category.

6.1.7 Financial Tools for Independence and Inclusion

Intermediary A2’s work across both Kumasi and Accra highlighted a specific design need: tools that support women’s financial independence and separation of business and household finances. Features that help women track business revenue distinctly from household expenditure, manage customer credit, set savings goals, and generate simple financial summaries were consistently valued by participants. Intermediary E3’s work with Solution S8 addressed a related challenge: peer-based credit access for women entrepreneurs in the Lower Manya, Asuogyaman, and Yilo Krobo districts.

- Include tools for savings goals, expense tracking, customer credit management, and digital receipts.
- Provide clear separation between business and household finance tracking.

- Consider P2P and community credit mechanisms alongside individual savings tools.

6.1.8 Including Women in the Design Process

Involving women as designers rather than only as testers is a logistical decision as much as a methodological one. In practice it means holding ideation sessions where and when women can actually attend: close to the market or workplace rather than at a hub, in short blocks that fit around trading hours and caregiving, and in the language participants think in. It means recruiting through structures they already belong to, such as trade associations, savings groups and Farmer-Based Organizations, which proved more reliable across this project than open calls. It also means giving participants something concrete to react to, since a rough prototype, a sketch or a role-played service draws sharper input than an abstract question about what someone would like. Where budget allows, bring the same women back across several rounds so that ideas can be developed with them rather than only collected from them.

See Section [3.4](#) for the principle behind this and the intermediary recommendations that produced it.

6.1.9 Visual Design Guidelines

Colour. Avoid gender stereotypes in colour choices: pink or pastel palettes to signal a product is “for women” are reductive and often counterproductive in professional business contexts. Colour choices should be brand-driven and context-specific, chosen for legibility and trustworthiness rather than gender signalling; high-contrast for outdoor readability on small screens; and consistent with local visual conventions where relevant, since colour associations vary by cultural context.

Typography. Use readable, trustworthy font faces; decorative or script fonts are inappropriate for functional interfaces. Minimum body text of 14–16pt for interfaces used on small or older devices. Ensure adequate line spacing to prevent text density from becoming a barrier for lower-literacy users. Label all interactive elements with text, not just icons.

Imagery. Use diverse representation: avoid imagery that reflects only one demographic, age group, or business type. Show realistic business scenarios rather than overly staged or aspirational stock imagery. Reflect the actual environments users work in: markets, workshops, farms, and home-based businesses. Where illustrating digital interfaces, ensure the illustrations match the actual product experience.

6.1.10 Cultural and Global Context

This guide is grounded in the Ghanaian context, but the principles it contains have relevance beyond Ghana. For practitioners adapting this guidance elsewhere:

- Infrastructure limitations differ by region. Connectivity, electricity reliability, and device type all vary. Assessment of local conditions must precede design decisions.

- Language accessibility requirements change by geography. What works in Greater Accra may not work in Tamale or in a different country entirely. Always research local language landscapes before deciding on language strategy.
- Cultural norms around money, privacy, and community obligation vary significantly. Design patterns that leverage community trust in one context may not map onto another.
- Gender dynamics are shaped by specific local, cultural, and legal contexts. Intersectional analysis of who the actual users are is always the starting point, not an assumption borrowed from another setting.

Where this guide is applied outside Ghana, treat it as a framework for asking the right questions rather than a set of answers that transfer without adjustment.

6.1.11 Awareness Setting for Investment Readiness

A recurring adoption barrier sits outside the interface. Many informal entrepreneurs do not yet treat digital tools as a business expense to plan and save for, so a subscription fee that is small in absolute terms can still fail if it arrives as a single unexpected cost, or before the tool has proven its value. Helping users understand that setting money aside for a proven tool is an investment in the business, not a loss, is part of designing for adoption.

The project's willingness-to-pay evidence points to what works. Solution P5's 5% platform fee tested well because participants compared it to the 40–60% interest they were already paying informal lenders. Solution P2's farmers accepted payment only after the service had been delivered. And once participants trusted a simple tool, they asked for savings-related features themselves (Solutions P5 and P7). The willingness to invest exists; it follows demonstrated value and familiar financial framing. Solution P11's validation with more than five hundred women farmers then strengthened this evidence at scale: a micro-pension component attracted stronger interest than the team had anticipated, farmers valued being able to contribute small amounts gradually through USSD while building long-term security, and several began contributing during the validation exercise itself. Willingness to commit rose when benefits were clearly explained and services were bundled, and fell where insurance and pension concepts were unfamiliar, which made agent-assisted onboarding and repeated community education sessions part of the product rather than an optional extra.

In practice:

- Anchor the price to a cost the user already tolerates (interest paid, stock lost, travel time), not to an abstract subscription rate.
- Sequence payment after value. Free first uses, pay-after-service, and escrow models proved more effective than upfront fees at any price level.
- Align payment amounts and timing with income patterns. Small mobile money instalments timed to market days are easier to plan for than monthly lump sums.
- Work through trusted community structures such as VSLAs, which already carry a disciplined savings culture, to build the habit of saving toward business tools.

- Make the return visible. A simple statement of what the tool saved or earned the user in a given period does more for renewal than any reminder message.

Where a team lacks its own evidence on this point, treat it as a research question in its own right: add willingness-to-pay and savings-behaviour questions to the screening and observation instruments in Sections [6.5](#) and [6.6](#) before settling on a pricing model.

6.2 Functional Design Patterns

These patterns are not prescriptive; they represent areas where thoughtful design creates disproportionate value for this user group. They should inform product prioritization rather than dictate feature lists.

Payment integration. Mobile money is the dominant financial infrastructure for women micro-entrepreneurs in Ghana. A digital business tool that does not integrate with MTN Mobile Money, Vodafone Cash, or AirtelTigo Money is operating outside the financial ecosystem participants actually use. Integrate at the point of transaction, not as a separate export step; support multiple mobile money operators, since preferences vary by region; display transaction confirmations in local language; allow offline payment recording with sync when connectivity returns. Intermediary N4 specifically recommended mobile money integration for Solution S5 platform, noting that trust and convenience were both served by connecting loyalty features to payment infrastructure women already trusted.

Social media and customer connectivity. Many women micro-entrepreneurs already use WhatsApp and Facebook for business communication, customer outreach, and order management. Allow sharing of receipts, product listings, and promotions via WhatsApp; support community features for group sharing, peer referral, and business visibility within trusted networks; provide templates for customer communication features, since blank text fields are a barrier while guided templates are a bridge. Intermediary A2 found that women who could share sales records and product updates through familiar channels were more likely to use a business management tool consistently.

Booking, scheduling, and service management. Service-based businesses (hairdressers, tailors, caterers, coaches) have specific scheduling and customer management needs that product-focused tools often do not address. Booking and appointment management features are high-value; reminders to both provider and customer reduce no-shows; simple customer record management (contact details, order history, outstanding credit) is consistently requested.

Content creation and business communication. Women entrepreneurs increasingly need to create promotional content, course materials, and service descriptions for digital channels. Ready-made templates for common business communication tasks reduce cognitive load; simple image and text creation tools that integrate with social sharing are valued for

promotional purposes. Intermediary N4's bulk SMS testing revealed that pre-written message templates with simple customization were the feature participants were most enthusiastic about.

Onboarding designed for gradual scaling. The best onboarding experiences in the project were those that started small: a participant who successfully completes one or two transactions on their first day is far more likely to return than one presented with the full feature set immediately. Introduce features progressively; provide sector-specific examples during onboarding (a food trader should see food-trading examples, not generic business terminology); allow users to reach a meaningful first success within the first five minutes of use. Intermediary E1 recommended breaking down functionality with step-by-step demonstrations using real-life business examples, building confidence before complexity.

6.3 Sample User Personas

These personas are composite profiles developed from participant data collected across the project. They are not based on any single individual. They illustrate the range of user profiles that gender-responsive digital solutions in peri-urban Ghana need to serve. The goal is not a single “typical female user” but a set of multi-dimensional profiles reflecting genuine diversity in age, geography, caregiving roles, literacy, digital experience, and sector.

Persona 1: Abena. Age 34. Location: Tamale, Northern Region. Business: second-hand clothing market stall. Education: Secondary school (SHS). Languages: Dagbani, Hausa, basic English. Digital access: shared Android smartphone, uses MTN Mobile Money daily. Goals: manage customer credit records, send payment reminders, build loyalty with repeat customers. Key barriers: English-only interfaces, poor outdoor readability, five-step transaction flows that interrupt customer service. *“I need something that works as fast as the customer pays. If it takes longer than writing it down, I stop using it.”*

Persona 2: Maame. Age 52. Location: Kumasi peri-urban, Ashanti Region. Business: food processing (palm oil, groundnut paste). Education: primary school only. Languages: Twi first language; no English. Digital access: her own basic Android phone; uses mobile money but no apps beyond messaging. Goals: track stock levels, know when to reorder, receive payments from buyers remotely. Key barriers: all-English interfaces, small font sizes, complex registration requiring email address. *“When they show me in Twi and big letters, I can do it. When it is small English, I give up.”*

Persona 3: Akosua. Age 28. Location: Lower Manya, Eastern Region. Business: hairdressing salon and braiding. Education: technical college graduate. Languages: Twi, Dangme, English. Digital access: personal smartphone, active on WhatsApp and Facebook for business. Goals: manage appointments, reduce no-shows, promote services to new customers, separate business income from personal spending. Key barriers: booking tools not integrated with WhatsApp, subscription fees she cannot predict month-to-month. *“I already manage my customers on WhatsApp. I want a tool that connects with that, not replaces it.”*

6.4 Gender-Responsive UI/UX Checklist

Use this checklist at every stage of product design and adaptation. It is not a substitute for field research, but a prompt for the questions that field research should be answering.

COMMON MISTAKES

Treating this checklist as a substitute for field testing, not a prompt for it. Every line below restates something the project learned by watching real participants use real products. Ticking every box in a meeting room, without testing any of it with actual users, produces a checklist that passed and a product that still fails in the field.

Accessibility and literacy

- Core user flows are available in at least one local language of the deployment area
- All interface text uses plain, concrete language with no unexplained jargon
- Font sizes are readable on small screens and in outdoor conditions (minimum 14pt body text)
- Colour contrast meets or exceeds WCAG 2.1 AA standards
- All interactive elements are labelled with text, not icons alone
- Audio guidance or voice features are available for key workflows
- Visual response scales and illustrated instructions are provided where text literacy is limited
- The product is tested on low-specification Android devices representative of the target user group

Operational realities

- Core functions work without an active internet connection
- Data synchronizes automatically when connectivity is restored
- Connection status is visible within the app
- Common actions can be completed in no more than two to three steps
- Partial task completion is saved if the user is interrupted
- Data consumption is minimized for slow connections and limited data plans
- The product performs acceptably on entry-level smartphones from 2019 onwards

Trust and safety

- Pricing and subscription terms are displayed clearly before sign-up
- Privacy policy is explained in plain language during onboarding
- PIN-protected or password-protected access to financial or personal data is available

- Quick logout or session lock is accessible within the main navigation
- Security features are communicated within the product, not only in documentation
- Human support contact is visible and accessible from the main interface

Cultural and gender responsiveness

- Visual design avoids gender stereotypes (not using pink or pastel schemes to signal “female”)
- Imagery reflects diverse representation across age, geography, and business type
- Imagery shows realistic business environments rather than staged stock photography
- The onboarding flow uses sector-specific examples relevant to the target user’s actual business
- Community and social features are considered alongside individual transaction features
- Shared device use has been accounted for in the privacy and profile architecture
- Women were involved in ideation or co-design activities, not only in testing
- The product has been tested with differently-abled participants

6.5 Participant Screening Template

This template can be adapted for telephone or in-person screening. Translate into local language before use. Use as a guide for conversation, not a rigid questionnaire.

- Business type and main products/services
- How long the business has been running
- Primary business location (market, shop, home-based, mobile)
- Whether they own or have regular access to a smartphone
- Which mobile money service they use (if any)
- Whether they currently use any apps for business (and which ones)
- Preferred language for the session (offer all available options)
- Availability and preferred session time
- Whether they have any accessibility requirements (vision, hearing, mobility)
- Whether they are comfortable being observed and having their session notes recorded (with anonymisation)

After screening, confirm: session date, time, and location; what to bring, if anything; that the session is about improving the product, not testing the participant; that participation is voluntary and they can stop at any time; and that their name will not appear in any reports or publications.

6.6 Usability Testing Observation Sheet

Complete one sheet per participant per session. Record observations in real time. Do not interpret during the session; record what you see and hear.

Session details: Participant reference number (do not use name) | Date and location | Facilitator name | Note-taker (if separate) | Solution being tested | Session start and end time.

Task observation log. For each task (use one block per task, typically 3–4 tasks per session), record: task name; completion status (yes / with help / no); time taken (approximate); observed behaviours (hesitations, workarounds, errors); participant quotes.

Post-task questions: What was easy about this task? What was difficult? What would you change if you could? Would you use this feature regularly, and why or why not? Overall impression of the session, in the participant's own words.

Facilitator observations: Note anything not captured in the task log: participant affect, confidence changes, moments of engagement or disengagement, environmental factors.

6.7 Ghana-Specific Localization Resources

Language and translation by region. Upper East Region: Dagbani, Mampruli, Gonja (Intermediary N1, Tamale-based operations; see Section 1.3's note on N1's region coding). Northern Region: Dagbani, Dagban, Mampruli, Gonja, Hausa (relevant intermediaries: Intermediary N4, Intermediary N3, Intermediary N2). Eastern Region: Twi, Dangme, Ewe, English (Intermediary E3, Intermediary E1, Intermediary E2). Ashanti Region: Twi, English (Intermediary A2, Intermediary A1, Intermediary A3, Intermediary A4, Intermediary A5). Greater Accra: Twi, Ga, Ewe, Hausa, English (Intermediary A2, secondary testing site).

Payment standards and mobile money integration. Mobile money is the dominant payment infrastructure for this user group. MTN Mobile Money (MoMo): largest network, available nationwide, highest usage among project participants. Vodafone Cash: significant presence in Southern Ghana. AirtelTigo Money: relevant in Northern and rural areas. Integration standard: the Ghana Interbank Payment and Settlement Systems (GhIPSS) provides the interoperability layer between mobile money providers; solutions should support GhIPSS-compliant APIs to enable cross-network transactions.

Cultural symbols and visual conventions. Adinkra symbols are widely recognized in Ghanaian visual culture and can be used purposefully in interface design, with appropriate understanding of their meanings. The colour green carries positive financial associations in some contexts (linked to savings and growth). Photographs and illustrations should reflect actual market and business environments; rural and peri-urban settings are underrepresented in stock imagery libraries and may require original photography.

Connectivity and infrastructure data sources. National Communications Authority (NCA) Ghana: quarterly statistics on mobile penetration and internet access. GSMA Intelligence: regional mobile data including gender gap analysis. Ghana Statistical Service: demographic and socioeconomic data from the 2021 Population and Housing Census. World Bank Open Data: economic indicators and financial inclusion data.

UX/UI: See Section [2.1](#).

Validation: See Field Validation above.

6.8 Ready-to-Use Templates and Downloadable Resources

The templates in Sections [6.5](#) and [6.6](#) are reproduced below as fillable tables; copy either table into a blank document, spreadsheet, or form tool and use it directly in the field. Translate into local language before use.

Participant Screening Template (fillable)

QUESTION	RESPONSE
Business type and main products/services	
How long the business has been running	
Primary business location (market, shop, home-based, mobile)	
Owns or has regular access to a smartphone (Y/N)	
Mobile money service used, if any	
Currently uses any apps for business (Y/N; which ones)	
Preferred language for the session	
Availability and preferred session time	
Accessibility requirements (vision, hearing, mobility)	
Comfortable being observed and recorded, with anonymisation (Y/N)	

Usability Testing Observation Sheet (fillable)

Session details:

FIELD	ENTRY
Participant reference number (not name)	
Date and location	
Facilitator name	
Note-taker (if separate)	
Solution being tested	
Session start / end time	

Task observation log (use one row per task, typically 3–4 tasks per session):

TASK NAME	COMPLETION (YES / WITH HELP / NO)	TIME TAKEN	OBSERVED BEHAVIOURS	PARTICIPANT QUOTES

External Template Galleries

For teams that want a cloud-based, shareable version of these forms, the Nielsen Norman Group's free UX templates and guides offer downloadable usability-test plans, screener scripts, and reporting templates that can be adapted for this context. The **Field Testing Guide** used across this project's own Phase 2 validation exercises (covering Assumption Framing, Problem Discovery, Solution/MVP Testing, and Business Model Testing, each with strong- and weak-signal indicators) is available as a companion document from the project team and is the most directly applicable structured template for any team continuing or replicating this project's field validation process.

07

Recommendations

What to do differently, by role.

The following recommendations consolidate the most consistent findings from across the project's intermediary cohort. They synthesize the direct input of all participating organizations, organized by the audience most likely to act on each recommendation. Numbering within each list reflects the order intermediaries most commonly raised each point in their submissions, not a formal priority ranking. Where this guide does intend a strict priority order (for example, which usability issues to fix first), it says so explicitly, as in Section [4](#), Part A, Step 3.

7.1 For Designers and Solution Providers

- Test in the field, not the office. Lab or hub testing will not surface connectivity issues, environmental distractions, shared-device dynamics, or literacy barriers that appear in real market environments.
- Prioritize language first. No other single improvement has as much impact on accessibility as making the core user journey available in the local language of the deployment region.
- Offline functionality is not optional for peri-urban and rural deployment. Use local storage with background synchronization so that core functions work without an active internet connection.
- Keep common actions fast. Two to three steps maximum for the most frequent tasks. Test this explicitly with real users under realistic conditions.
- Invest in human support. An app users know they can call for help is used more than one that leaves them on their own.
- Design for interruption. Real business environments are noisy and unpredictable. Tasks must be saveable and resumable without data loss.
- Provide ready-made templates and pre-built structures for common business tasks.

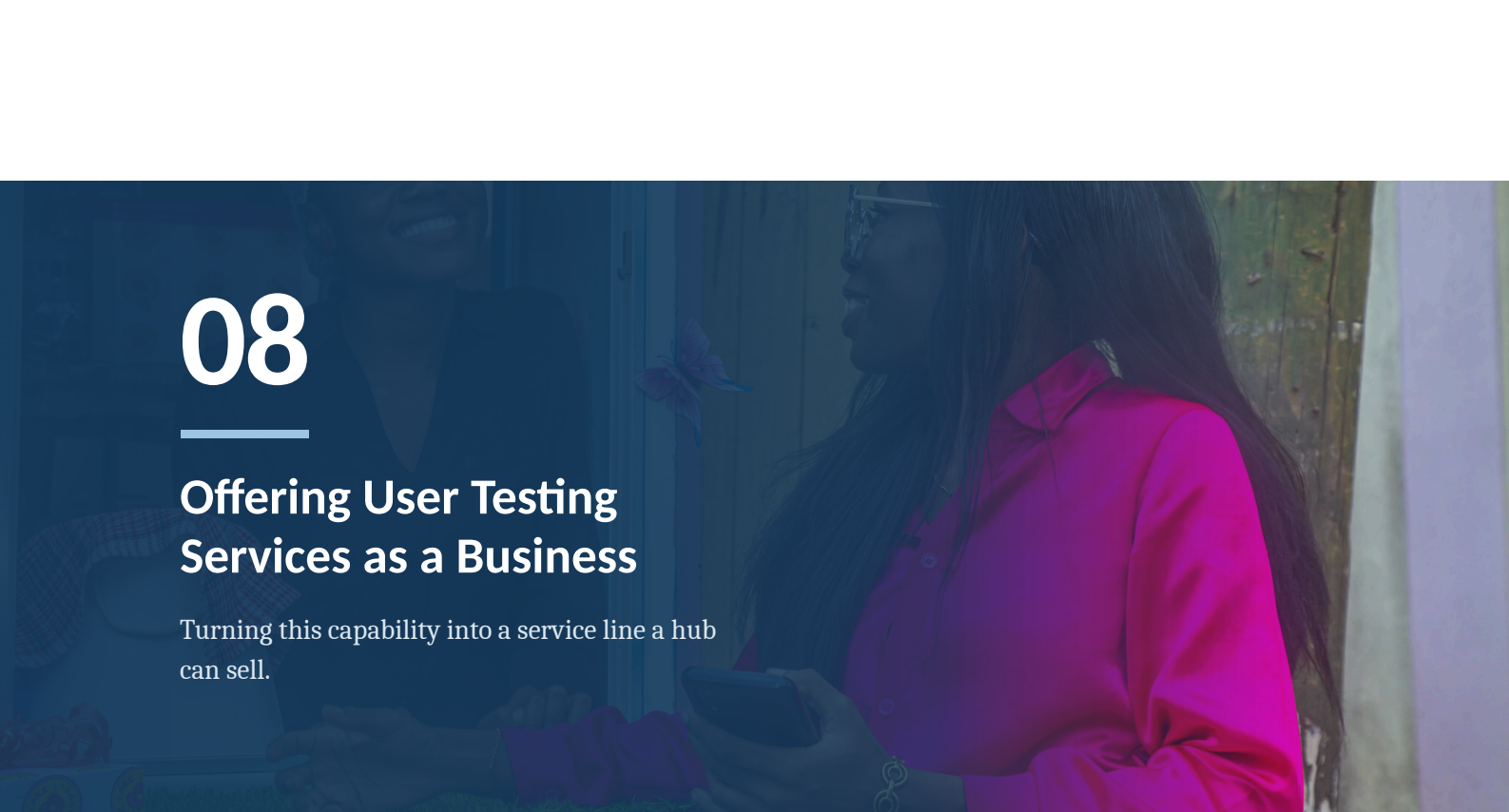
- Fix bugs before anything else. Unresolved bugs destroy trust among first-time users faster than any other issue.
- Include women in ideation, not just testing. Participatory design workshops involving women micro-entrepreneurs produce fundamentally better products.
- Consider differently-abled users explicitly. Gender-responsive design must go beyond the average user profile.
- Avoid gender stereotypes in visual design. Choose colour, typography, and imagery based on context, readability, and trust, not gender assumptions.
- Budget for user research as a core development cost, not an optional extra. Structured testing takes money and time; providers who plan and fund it from the start are the ones able to act on the findings it produces.

7.2 For Intermediaries Running User Research

- Invest in product knowledge before fieldwork. Intermediaries who understand the product deeply produce better findings.
- Recruit through trusted community channels. Market queens, savings groups, and peer referral consistently outperform cold outreach.
- Recruit more participants than the minimum. Attrition is predictable, so build the buffer in from the start.
- Recruit for diversity deliberately. Over-relying on literate participants hides significant usability gaps.
- Design instruments for the participant, not the reporting template. Simple, conversational, visually supported tools produce better data.
- Keep sessions short and focused. Fifteen to twenty minutes for questionnaires combined with task testing.
- Maintain the same participant cohort through the full project lifecycle, and budget for the participant incentives that make this possible. People are being asked to stay engaged through research, iteration, and validation rounds; modest, transparent compensation for time and transport is often what keeps a cohort together to the end.
- Link every recommendation to specific user evidence. Observation notes and direct quotes are what move solution providers to act.
- Document what did not work. The record of failure is as valuable as the record of success.
- Escalate early when solution providers are unresponsive.
- Create structured peer-exchange moments for the solution providers you support. The virtual demo day held ahead of the project's final pitch event pushed several teams to improve further; seeing what colleagues were building, and hearing their feedback, did work that individual coaching alone did not. This is a low-cost practice any intermediary can continue with its own cohort.

7.3 For Project and Programme Managers

- Assess solution provider readiness before selection. Evaluate development capacity, not just intent.
- Build in enough time for co-creation to be substantive.
- Make training attendance flexible.
- Support intermediary organizations to continue this work beyond the project period; Section 8 shows how that can take the form of a paid service line.
- Conduct post-project evaluation surveys that include solution providers.
- Build solution-provider accountability into project design.
- Monitor adoption after the project ends.
- Provide dedicated software engineering support when incubating non-technical founders. In Phase 2, this lowered the entry barrier for non-technical teams, who could build a proof of concept and keep momentum while still searching for technical team members. Intermediaries rated it among the most valuable design choices of the programme.



08

Offering User Testing Services as a Business

Turning this capability into a service line a hub can sell.

This section reflects the project team's observation that there is strong potential to replicate the approach as a service integrated into the portfolios of innovation hubs and business support intermediaries. It provides guidance on how these organizations can sustainably offer user testing as a commercial service beyond the lifetime of the project. The section is intended for hub leadership and programme managers who are considering whether, and how, to package and deliver user testing as a fee-based service.

8.1 Why This Is a Viable Service Line

The Project itself is the market signal. GIZ supported twelve Ghanaian organizations to build exactly this capability: structured user research, gender-sensitive facilitation, and evidence-based co-creation support, because Ghanaian solution providers do not have it in-house and cannot easily buy it elsewhere. That gap does not close when the project ends. If anything, the pipeline of early-stage digital solutions targeting informal-sector and women-led businesses in Ghana is growing, not shrinking, and most of those teams will face the same problem the solution providers faced: they are building for users they do not understand well enough, and they know it.

Two distinct buyer segments emerged from the project cohort, and a hub packaging this service should recognize both:

- **Under-resourced early-stage teams** (similar to several Phase 2 ideation teams) who need low-cost, fast testing to validate direction before they have built much: price-sensitive, but a natural funnel from a hub's own incubation or accelerator programme.
- **Better-resourced solution providers** (similar to several Phase 1 partners) who have a live product and need ongoing, rigorous testing and co-creation support to keep improving it: able to pay closer to market rate, but only if the hub can demonstrate it delivers findings a provider can act on, not just a report.

8.2 What a Hub Needs to Offer This Credibly

- **A trained facilitator, not just a UX-literate staff member.** Section [4](#), Part A is effectively a training curriculum: user-centred design methods, task-based testing, neutral observation, gender-sensitive facilitation. A hub does not need a full-time UX team to start. It needs at least one person who has been through this discipline properly, plus a documented process others can follow.
- **A genuine community recruitment network.** This is the project's cohort's actual competitive advantage over a generic UX consultancy, and it is worth naming explicitly to clients: hubs with real relationships with market queens, savings groups, and trade associations can recruit representative, honest participants faster and more credibly than an outside firm can. This is the thing to sell, not just “we can run usability tests.”
- **A reusable toolkit.** The screening template, observation sheet, and checklist in Section [6](#) are ready to use as-is. Standing up this service does not require building research instruments from scratch.
- **A track record.** Even one well-documented case study (ideally one the hub ran itself, not just this guide's project examples) does more to sell the service than any amount of description. Section [8.5](#) below explains how to build that case study.

8.3 Packaging the Service

There is no single right model. Four approaches showed up, implicitly or explicitly, across how the project itself was structured and how comparable hub-building efforts in Ghana (including incubation programmes built from scratch, such as the RUG Hub/RUG Incubation pattern) have introduced new service lines:

MODEL	HOW IT WORKS	BEST FOR	WATCH OUT FOR
Fixed-scope sprint	A defined engagement (e.g. one round of task-based testing with 8–10 participants, a findings report, and a prioritized recommendation list), priced and scoped upfront	First paying clients; building a portfolio quickly	Underscoping the recruitment effort, which is usually the most time-intensive part
Retainer / ongoing co-creation partner	The hub embeds with a solution provider across multiple test-adapt-retest cycles, mirroring Section 4's Step 1–3 loop	Better-resourced providers who need continuous improvement, not a one-off report	Requires the provider to actually have development capacity to act on findings (see Section 4, Part A, “When providers are reluctant, slow, or unable to change”); assess this before signing
Bundled into an existing incubation or accelerator programme	User testing is built into the cohort curriculum and fee structure rather than sold separately	Hubs that already run incubation programmes and want to raise the quality bar for graduating teams	Risk of treating it as a checkbox workshop rather than a real testing cycle with follow-through
Grant- or donor-subsidized pilot cohort	A funder (GIZ, another donor, or GHN itself) underwrites the first few engagements so the hub can build case studies without asking early clients to pay full price	Hubs with no track record yet	Needs a deliberate transition plan to paid engagements; subsidized pilots that never convert to revenue are not a service line, they are a grant

Most hubs will want to start with a subsidized or discounted pilot, convert that into one or two fixed-scope sprints with documented outcomes, and only then consider a retainer model or bundling into a flagship programme.

8.4 Pricing Considerations in the Ghana Context

This guide does not set specific price points, since pricing structure will vary by hub capacity, region, and client type. Three considerations should shape whatever a hub decides:

- Most likely early clients, such as informal-sector-facing startups, NGO-backed pilots, other GHN-network solution providers, are cash-constrained in the same way the end users this guide describes are. Flat, transparent, pay-as-you-go pricing per engagement will land better than a subscription or retainer proposal to a team that has never bought this service before.

- Segment pricing deliberately. A venture-backed or foreign-funded client can reasonably be charged closer to market consulting rates; an informal-sector-facing early-stage team from the hub’s own ecosystem may need a reduced or in-kind-for-equity/data-sharing arrangement, at least initially.
- Use MoMo-based invoicing and payment where possible. It is the payment method the hub’s own future clients will find least friction in, consistent with Section 6.2’s evidence on payment infrastructure.
- Expect to innovate on the payment model itself. Solution providers rarely hold a budget line for user research, a point hub across the GHN network have raised consistently. Models that spread or defer the cost, such as instalment payments by mobile money, milestone-based billing tied to co-creation cycles, or bundling research into a wider incubation package, widen the pool of clients who can realistically say yes.

8.5 Demonstrating Return on Investment

The strongest sales tool a hub has is a concrete before-and-after case study, quantified wherever possible. Two examples from this guide illustrate the format to reuse: Intermediary N1’s Solution S1 password-simplification case (Section 4, Part A, Step 3) shows a specific, low-cost fix moving login from a common failure point to near-universal success in one testing cycle. Intermediary A1’s Solution S9 data (Section 5.1) shows the reverse: a concrete, previously-unknown failure rate (36.4% on a core task) that the provider could not have found without structured field testing.

A hub selling this service should be able to produce the same shape of evidence for its own first client within one engagement: a specific problem, a specific fix, a specific measured result. Generic claims about “improving usability” will not persuade a second client the way one real, numbered result will.

For early-stage clients there is a second return worth naming in the sales conversation: research participants are often a solution’s first adopters. A well-recruited cohort that has shaped a product through several rounds of testing is also a ready-made group of early customers and references once the product launches.

PRACTICAL TIP

Your first sales asset is your first client’s own data, not a slide deck. A hub does not need a polished portfolio to start selling this service; it needs one completed engagement written up the way Section 4, Part A, Step 3 documents co-creation (“User feedback indicated [issue]. This led us to recommend [specific change]. Result: [measured outcome].”). One real number beats ten generic claims.

8.6 Risks and What to Avoid

- **Do not sell testing without the follow-through capacity to support co-creation.** The project evidence is unambiguous (Section [4](#), Part A, “When providers are reluctant, slow, or unable to change”): testing that does not lead to acted-upon change produces no measurable benefit and damages the hub’s credibility with that client. If a hub cannot also coach a client through prioritizing and implementing findings, it should say so upfront and scope the engagement accordingly.
- **Assess client readiness before selling, the same way GFA had to assess solution-provider readiness before selection.** A client with no development capacity to act on findings is not a good first paying client, however willing they are to pay for a report.
- **Do not oversell before running at least one real pilot.** Credibility in this market is built on demonstrated results, not marketing language (see Section [8.5](#)).
- **Keep the evidence-based standard this guide holds itself to.** A hub’s commercial pitch should be as honest about what testing can and cannot tell a client as this guide is about the project’s own limitations (Section [5.7](#)).

09

Sustaining the Open-Source Guide

How the guide stays alive after the project ends.

9.1 What to Contribute

This guide is intended to be a living document. The field of digital inclusion for women entrepreneurs in sub-Saharan Africa is moving quickly. User behaviours are changing as smartphone penetration increases, the product landscape is evolving, and local language technology is improving. What holds true today about connectivity constraints may look different in a few years. The value of this guide over time depends on whether new practitioners engage with it, challenge it, and update it with their own evidence.

The most useful contributions are specific and evidence-based. If your project confirmed something in this guide, say so. If your findings contradict something, say that too, and explain the context: region, product type, and user demographics all matter. Priority contribution areas:

- New case examples from similar projects in Ghana or elsewhere in the region.
- Updated data on connectivity, smartphone ownership, or digital literacy patterns.
- Additional screening templates, observation guides, or evaluation frameworks adapted for specific contexts.
- Lessons from product types or user groups not covered in this edition.
- Additional Phase 2 solutions and later Phase 2 cohorts not yet reflected in Section [4](#), Part B and Section [5.6](#).
- Translations of this guide or its appendices into local languages.
- Design patterns or component libraries tested with this user population.
- Validation evidence from AI-enabled solutions, particularly speech interfaces in Ghanaian languages, as Solutions S13, P8, P9, and P10 mature (see Section [5.8](#)).

9.2 Contribution Guidelines

Contributions to this guide should be practical and based on documented field activity, not general design theory; clearly referenced, stating what project, context, and evidence base the contribution draws on; aligned with user-centred design principles and gender-responsive practice; and submitted with transparency about limitations, presenting context-specific findings as such.

Help improve the guide. Choose the route that best matches what you want to contribute. Each route is public and requires a free GitHub account.

1. **Report a correction.** Flag inaccurate or outdated information, unclear guidance, broken links, or editorial mistakes. Small, well-defined corrections are the easiest contributions to review.
2. **Share field evidence or a case study.** Add findings that confirm, challenge, or extend the guide. The submission asks about context, participants, methods, findings, sources, and limitations, so that future maintainers can assess it without extensive follow-up.
3. **Edit the guide directly.** Select *Edit this page* at the top of any guide page to propose a change to its Markdown source. GitHub will create a pull request; you cannot break the published guide by proposing an edit.
4. **Volunteer to help maintain the guide.** Practitioners and organizations can offer review, accessibility, research, translation, or technical stewardship.

Before opening an issue or pull request, search the [existing issues](#) to see whether the same contribution is already being discussed. All four routes start from the guide repository at github.com/joekay-kiipo/os-guide-site.

MAINTAINED ON A BEST-EFFORT BASIS

This guide is an open-source, community-maintained resource. Contributions are welcome, but response and review times depend on whether an active maintainer is available. Submission does not guarantee inclusion. If the project becomes inactive, anyone may continue the work through a fork, or publish an adaptation under the licence in Section 9.5.

Two conditions apply. Contributions are published under the same CC BY 4.0 licence as the rest of the guide, so please send only material you are free to share on those terms. Anything describing a participant, a business, or a solution provider must already be anonymised, or must come with that person's permission: the standard set out in Section 5.7 applies to contributions exactly as it applied to the original research.

Contradictory findings are welcome. Findings that contradict this guide are among the most valuable contributions it can receive. If your evidence disagrees with something here, say so and explain the context: region, product type, and user demographics all matter.

9.3 Review and Governance Model

Ownership of this guide is intended to transfer from the GFA/GIZ Project team to a Ghana Hubs Network (GHN) open-source community as part of the project's sustainability plan. Until a standing maintainer group exists, the GitHub repository is the public source of truth for proposed changes, review discussion, accepted contributions, and project status.

Minor corrections may be merged by any active repository maintainer after a basic accuracy check. New research claims, statistics, case studies, or substantial changes should be supported by documented evidence and reviewed by someone with relevant subject knowledge before publication. Automated checks may verify links, formatting, and required submission fields, but they do not replace human judgement about evidence, safeguarding, or contextual accuracy.

Major updates should follow versioned releases; minor corrections and additions can be incorporated in interim updates. If no maintainer remains available, the repository should say so clearly and invite new maintainers. If long-term stewardship cannot be found, the repository may be archived as a read-only record while retaining the right for others to fork and adapt it under CC BY 4.0.

9.4 Knowledge-Sharing Platforms

This guide is intended to be distributed and built upon through a Git-based repository (GitHub or GitLab) for version-controlled collaborative editing; GHN community forums and practitioner networks focused on digital inclusion and gender-responsive design; workshops, learning sessions, and design clinics organized under GHN's knowledge-sharing agenda; and partner organization publications and distribution channels.

9.5 Licence

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9.6 Sustaining This as a Revenue Stream for GHN Members

Section 8 sets out how individual hubs can turn the capability this guide documents into a paid service. From a network-sustainability perspective, that is also what keeps this guide alive in practice: hubs that are actively running user testing engagements are the ones most likely to generate the new case studies, corrections, and Phase 2 content this guide will need (Section 9.1). Treat Section 8 and this section as two sides of the same goal: one hub's income, and the whole network's evidence base.

9.7 Applying the Guide Flexibly

No two projects are identical. The guidance here comes from a specific context: Ghana, 2023–2025, peri-urban and rural women micro-entrepreneurs. Before applying any recommendation elsewhere, ask what is the same and what is different. The principles tend to transfer; the specific implementation details often need adjustment.

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